Olist Brazilian E-Commerce Analytics

End-to-End Business Intelligence and Predictive Analytics Study



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Olist

Executive Summary

Olist Brazilian E-commerce Analytics: An End-to-End Business Intelligence and Predictive Analytics Study

This project presents an end-to-end analysis of the Olist Brazilian E-Commerce Public Dataset, with the aim of transforming transactional data into actionable business insights. The study examines sales performance, product and category contribution, customer purchasing behavior, seller revenue concentration, and delivery operations. It also explores the relationship between weather conditions and delivery delays and develops machine-learning models to identify orders with elevated late-delivery risk. The analytical results are consolidated into an interactive Power BI dashboard designed to support business monitoring and decision-making.

The project follows a multi-stage analytical workflow combining Python, SQL, external API integration, machine learning, and business intelligence. Python was used for data preparation, exploratory analysis, customer-level feature engineering, and RFM segmentation. MySQL was used to structure and query the data, calculate business metrics, and investigate performance across customers, products, sellers, locations, and time. Historical weather observations were retrieved through the Open-Meteo API and matched to a subset of orders. Logistic Regression and Random Forest models were then developed to examine the feasibility of identifying late-delivery risk using order and geographic characteristics. Finally, Power BI was used to present the main business indicators and analytical findings through dedicated dashboard pages.

The dataset contains **99,441 orders**, of which **96,478 were delivered**, corresponding to a delivery completion rate of **97.02%**. Total product revenue across all order statuses amounted to approximately **€13.59 million**, while product revenue associated with delivered orders reached approximately **€13.22 million**. These measures are reported separately because they represent different order populations. The monthly revenue analysis showed an overall increase during the observed period, with November 2017 recording the highest monthly revenue in the analysis, at approximately **€988,000**.

Product-level analysis identified Health & Beauty, Watches & Gifts, and Bed, Bath & Table among the leading categories by delivered product revenue. Bed, Bath & Table also recorded the highest unit volume among the leading categories examined. These results demonstrate the value of assessing category performance through both monetary contribution and sales volume, rather than relying on a single measure.

Customer analysis revealed a substantial difference between the number of customers who purchased once and those who made repeat purchases. In the SQL customer analysis, repeat customers represented approximately **3%** of customers, while their average order value was approximately **€260.05**, compared with **€137.96** for one-time customers. The RFM analysis further distinguished customer groups according to recency, frequency, and monetary value. Potential Loyalists represented approximately **40.06%** of the RFM population and contributed

approximately **40.95%** of the revenue attributed to the analyzed segments. The results provide a basis for investigating customer retention and engagement opportunities, while also showing why high spending should not automatically be interpreted as customer loyalty.

Delivery analysis found that **6,535 orders** were late among delivered orders with both the actual and estimated delivery dates available, producing a late-delivery rate of **6.77%** for that defined population. Delivery performance varied across geographic areas, although rates in locations with relatively few orders require cautious interpretation. The weather analysis used a final matched sample of **40,621 orders**. Within this sample, the late-delivery rate was approximately **5.89%** on dates without rain and **7.41%** on rainy dates. This difference represents an observed association rather than evidence that rainfall caused delivery delays. Weather recorded on the actual delivery date was therefore treated as an exploratory analytical variable and excluded from the pre-delivery prediction features.

The predictive modeling phase compared Logistic Regression and Random Forest for late-delivery classification. The Random Forest achieved an ROC-AUC of approximately **0.774**, compared with approximately **0.706** for Logistic Regression. At the selected illustrative probability threshold of 0.50, the Random Forest identified approximately **62.9%** of late orders, with a late-order precision of approximately **18.0%**. This operating point illustrates a trade-off between identifying late orders and generating false alerts. The model is consequently interpreted as a potential prioritization aid for operational review, not as a definitive prediction of delivery outcomes.

Seller analysis showed that revenue was concentrated among a subset of sellers: the top 10% accounted for approximately **41.10%** of seller revenue, while the top 20% accounted for approximately **56.62%**. This concentration provides a reason to examine seller contribution alongside operational indicators, rather than treating revenue share alone as a complete measure of seller performance or risk.

The project's main deliverable is an interactive Power BI report covering Sales Overview, Product & Category Intelligence, Orders & Delivery, Customer Overview, and Seller Overview. The dashboard brings together key performance indicators, category and geographic comparisons, customer segmentation, and seller analysis in a single reporting environment. It complements the underlying Python and SQL work by making selected findings accessible for business exploration.

Overall, the study demonstrates how multiple analytical methods can be combined to examine an e-commerce business from commercial, customer, seller, and operational perspectives. The results identify customer retention, delivery monitoring, and seller concentration as areas for further investigation. They also illustrate the importance of defining metric populations carefully, distinguishing association from causation, and evaluating predictive models according to their intended operational use. Because the analysis is based on historical data and does not include all costs or operational variables, its findings should be interpreted as an analytical foundation for further investigation and testing rather than as a complete assessment of current business performance.

Table of Contents

Preliminary pages....0
Cover Page....0
Executive Summary....1
Table of Contents....3
List of Figures....8
List of Tables....9
List of Abbreviations....11
1. Introduction....12
1.1 Project Background and Context....12
1.2 Problem Statement....12
1.3 Project Objectives and Research Questions....13
1.4 Scope of the Study....14
1.5 Report Structure....14
2. Dataset and Analytical Framework....15
2.1 Olist Dataset Overview....15
2.2 Data Sources and Database Structure....16
2.3 Data Preparation and Quality Assessment....17
2.4 Analytical Workflow and Tools....19
2.4.1 Data Preparation and Exploratory Analysis with Python....19
2.4.2 Relational Analysis with MySQL....19
2.4.3 External Data Enrichment through the Open-Meteo API....19
2.4.4 Predictive Modeling with Machine Learning....20
2.4.5 Business Intelligence and Visualization with Power BI....20
2.4.6 Integration of Analytical Results....20
2.5 KPI Definitions and Measurement Scope....21
2.5.1 Sales and Order KPIs....21
2.5.2 Product and Category KPIs....22
2.5.3 Customer KPIs....22
2.5.4 Delivery KPIs....22
2.5.5 Predictive Modeling Metrics....23
2.5.6 Interpretation Across Different Analytical Populations....23
3.1 Overall Sales and Order Performance....24
3.2 Revenue Evolution Over Time....26
3.3 Product and Category Performance....27
3.3.1 Revenue Contribution by Product Category....27
3.3.2 Sales Volume and Product Price....28
3.3.3 Implications for Business Analysis....28
3.4 Geographic and Seller Performance....29
3.4.1 Geographic Distribution of Customer Demand....29

3.4.2 Seller Revenue and Geographic Activity..... 30
3.4.3 Seller Revenue Concentration..... 30
3.4.4 Business Implications..... 31
4. Customer Analytics and Segmentation..... 34
4.1 Customer Base and Purchase Frequency..... 34
4.1.1 Overview of the Customer Base..... 34
4.1.2 Purchase Frequency and Repeat Customers..... 34
4.1.3 Purchase Frequency and Customer Value..... 35
4.1.4 Interpretation..... 35
4.2 Customer Spending and Monetary Value..... 35
4.2.1 Average Order Value..... 35
4.2.2 Customer Spending Distribution..... 36
4.2.3 Revenue Concentration Among Customers..... 37
4.2.4 Interpretation..... 37
4.3 RFM Methodology and Customer Segments..... 38
4.3.1 RFM Dimensions..... 38
4.3.2 Scoring Approach..... 38
4.3.3 Customer Segment Distribution..... 39
4.3.4 Interpretation and Limitations..... 40
4.4 Segment Revenue Contribution and Retention Implications..... 40
4.4.1 Revenue Contribution by Segment..... 41
4.4.2 Comparing Customer Share and Revenue Share..... 41
4.4.3 Retention Implications..... 41
4.4.4 Summary..... 42
5. Delivery and Weather Analysis..... 42
5.1 Delivery Performance Overview..... 42
5.1.1 Delivery Time..... 42
5.1.2 Delivery Status Distribution..... 42
5.1.3 Interpreting Delivery Outcomes..... 43
5.1.4 Data Considerations..... 44
5.2 Late Deliveries Across Time and Geography..... 44
5.2.1 Monthly Late-Delivery Patterns..... 44
5.2.2 Geographic Differences in Delivery Risk..... 45
5.2.3 Interpretation..... 46
5.3 Weather Data Integration and Analysis..... 46
5.3.1 Weather Data Collection and Integration..... 46
5.3.2 Weather Conditions and Late-Delivery Rates..... 47
5.3.3 Interpretation and Limitations..... 48
5.3.4 Summary..... 48
5.4 Integrated Interpretation of Delivery and Weather Findings..... 48
6. Predictive Modeling..... 49

6.1 Modeling Objective and Target Definition..... 49
6.1.1 Modeling Objective..... 49
6.1.2 Target Definition..... 49
6.1.3 Modeling Population and Scope..... 50
6.2 Feature Selection and Modeling Approach..... 50
6.2.1 Selected Features..... 50
6.2.2 Train-Test Split..... 51
6.2.3 Models Evaluated..... 52
6.2.4 Feature Availability and Interpretation..... 52
6.3 Model Evaluation and Performance..... 52
6.3.1 Evaluation Metrics..... 52
6.3.2 Logistic Regression and Random Forest Results..... 53
6.3.3 Interpreting the Results..... 54
6.3.4 Summary..... 54
6.4 Classification Threshold and Feature Importance..... 54
6.4.1 Classification Threshold Analysis..... 54
Table 14. Classification Threshold Analysis..... 55
6.4.2 Selected Threshold..... 55
6.4.3 Feature Importance..... 55
6.4.4 Interpretation and Limitations..... 56
6.5 Predictive Modeling Summary and Limitations..... 56
6.5.1 Main Limitations..... 56
6.5.2 Overall Contribution..... 57
7. Power BI Dashboard and Business Intelligence..... 57
7.1 Data Model and Dashboard Design..... 57
7.1.1 Data Model..... 57
7.1.2 Dashboard Structure..... 58
7.1.3 Measures and Interactive Analysis..... 58
7.1.4 Design Considerations..... 58
7.2 Sales Overview Dashboard..... 59
7.2.1 Key Performance Indicators..... 59
7.2.2 Revenue Evolution..... 60
7.2.3 Geographic Perspective..... 60
7.2.4 Dashboard Role..... 60
7.3 Product & Category Intelligence Dashboard..... 61
7.3.1 Key Performance Indicators..... 61
7.3.2 Category Performance..... 61
7.3.3 Product-Level Analysis..... 62
7.3.4 Dashboard Role..... 62
7.4 Orders & Delivery Dashboard..... 63
7.4.1 Delivery Performance Indicators..... 63

7.4.2 Temporal Delivery Patterns....63
7.4.3 Geographic Delivery Risk....63
7.4.4 Dashboard Role....63
7.5 Customer Overview Dashboard....64
7.5.1 Customer Base and Repeat Purchasing....64
7.5.2 RFM Customer Segmentation....65
7.5.3 Customer Value....65
7.5.4 Geographic Customer Distribution....65
7.5.5 Dashboard Role....65
7.6 Seller Overview Dashboard....66
7.6.1 Seller Performance Indicators....66
7.6.2 Geographic Distribution of Sellers....66
7.6.3 Seller Revenue Analysis....67
7.6.4 Dashboard Role....67
7.7 Dashboard Usability and Analytical Limitations....68
8. Integrated Insights and Business Recommendations....70
8.1 Integrated Analytical Findings....70
8.2 Business Recommendations....72
8.2.1 Strengthen Customer Retention....72
8.2.2 Establish Delivery-Risk Monitoring....72
8.2.3 Monitor Delivery Performance by Geography and Time....72
8.2.4 Manage Seller Concentration....73
8.2.5 Use Category-Level Performance for Product Decisions....73
8.2.6 Establish a KPI Monitoring Framework....73
8.3 Implementation Priorities and Monitoring Framework....75
9. Conclusion and Future Work....77
9.1 Conclusion....77
9.2 Limitations and Future Work....77
References....79
Appendix A — Dataset and Data Dictionary....80
A.1 Dataset Overview....80
A.2 Main Fields and Definitions....81
Orders....81
Order Items....82
Customers....83
Products....83
Sellers....84
A.3 Derived Analytical Tables....84
Appendix B — SQL Queries and KPI Definitions....85
B.1 SQL Analysis Overview....85
B.2 Main KPI Definitions....85

B.3 Revenue and Order Performance..... 86
B.4 Customer Purchasing Behavior.....87
B.5 Delivery Performance.....88
B.6 Seller Revenue Concentration..... 88
Appendix C — Python Workflow and Data Preparation.....88
C.1 Overview..... 88
C.2 Data Preparation.....89
C.3 Exploratory Data Analysis.....89
C.4 Customer Segmentation..... 89
C.5 Weather Data Integration.....90
C.6 Predictive Modeling.....90
C.7 Outputs.....91
Appendix D — Predictive Modeling Details and Evaluation.....91
D.1 Modeling Objective.....91
D.2 Dataset and Target Definition.....91
D.3 Predictive Features..... 92
D.4 Training and Evaluation Approach..... 93
D.5 Model Performance.....94
D.6 Classification Threshold Analysis.....94
D.7 Feature Importance.....95
D.8 Limitations.....96
Appendix E — Power BI Dashboard Documentation.....97
E.1 Dashboard Overview.....97
E.2 Dashboard Pages.....97
E.2.1 Sales Overview.....97
E.2.2 Product & Category Intelligence.....97
E.2.3 Orders & Delivery..... 98
E.2.4 Customer Overview.....98
E.2.5 Seller Overview..... 98
E.2.6 Navigation Page..... 98
E.3 Data Model and Relationships..... 98
E.4 Dashboard Interactivity.....99
E.5 Dashboard Scope and Limitations..... 100

List of Figures

Figure	Title
Figure 1	Relational data model used for the Olist analysis
Figure 2	End-to-end analytical workflow of the Olist project
Figure 3	Sales Overview dashboard and headline performance indicators
Figure 4	Monthly product revenue over the observed period
Figure 5	Delivered product revenue by category
Figure 6	Units sold by product category
Figure 7	Delivered orders by customer state
Figure 8	Seller revenue concentration
Figure 9	Seller geographic distribution
Figure 10	Average Customer Spend by Segment
Figure 11	Customer Segment Distribution
Figure 12	Order Status Distribution
Figure 13	Monthly Late Rate
Figure 14	Delivery Risk by State
Figure 15	Power BI Dashboards Navigation Page
Figure 16	Power BI Sales Overview Dashboard
Figure 17	Power BI Product & Category Overview Dashboard
Figure 18	Power BI Orders & delivery Dashboard
Figure 19	Power BI Customer Overview Dashboard
Figure 20	Power BI Seller Overview Dashboard

List of Tables

Table	Title
Table 1	Overview of the principal Plist dataset tables
Table 2	Analytical populations used in the project
Table 3	Core KPI definitions and scope
Table 4	Overall sales and performance
Table 5	Seller revenue analysis
Table 6	RFM Segmentation and Customer Characteristics
Table 7	Frequency of Number of Orders
Table 8	Delivery Status Distribution
Table 9	Delivery Performance Indicators
Table 10	Late-Delivery Rates by Weather Condition
Table 11	Late-Delivery Rates by Rain Intensity
Table 12	Predictive Modeling Dataset
Table 13	Predictive Model Performance Comparison
Table 14	Classification Threshold Analysis
Table 15	Business Area and Monitoring KPIs
Table 16	Implementation Priorities
Table 17	Dataset Tables Definitions
Table 18	Orders Table columns definitions
Table 19	Order_items Table columns definitions
Table 20	Customer Table columns definitions
Table 21	Products Table columns definitions
Table 22	Sellers Table columns definitions

Table 23	Derived Analytical Tables
Table 24	Main KPI definitions
Table 25	Revenue & order Performance
Table 26	Customer Purchasing Behavior
Table 27	Customer Segmentation
Table 28	Dataset & Target Definition
Table 29	Predictive Features Description
Table 30	Model Performance
Table 31	Classification Threshold Analysis
Table 32	Feature Importance
Table 33	Data Model & Relationships

List of Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
AOV	Average Order Value
BI	Business Intelligence
CSV	Comma-Separated Values
DAX	Data Analysis Expressions
EDA	Exploratory Data Analysis
KPI	Key Performance Indicator
ML	Machine Learning
RFM	Recency, Frequency, Monetary
SQL	Structured Query Language
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
SP	São Paulo
RJ	Rio de Janeiro
BA	Bahia
PB	Paraíba

1. Introduction

1.1 Project Background and Context

E-commerce platforms generate large volumes of transactional data through customer orders, product purchases, seller activity, and delivery operations. When properly structured and analyzed, these data can provide valuable information about business performance, customer behavior, and operational processes.

The growth of online commerce has increased the importance of understanding not only how much a business sells, but also how customers interact with the platform, which products contribute to sales, how performance varies across geographic areas, and whether orders are delivered within their estimated timeframes. These questions require an analytical approach that connects different dimensions of business activity rather than examining individual metrics in isolation.

This project focuses on the Olist Brazilian E-Commerce Public Dataset, which provides historical information about orders, customers, products, sellers, payments, and logistics-related events. Its relational structure makes it suitable for investigating e-commerce performance from multiple perspectives.

The project was developed as an end-to-end analytics study, combining data preparation, exploratory analysis, SQL-based business querying, external weather-data integration, predictive modeling, and interactive business intelligence reporting. By connecting these techniques within a single workflow, the study aims to demonstrate how transactional data can be transformed into structured findings that support business understanding and further investigation.

1.2 Problem Statement

E-commerce performance cannot be fully understood through total sales or order volume alone. A more comprehensive analysis must also consider the composition of revenue, customer purchasing patterns, seller contribution, and the reliability of delivery operations.

The central problem addressed in this project is how to transform the Olist dataset into an integrated analytical view of business and operational performance. This involves identifying the factors and patterns that characterize sales and customer behavior, examining delivery outcomes across time and geography, and evaluating whether historical order information can help identify orders with elevated late-delivery risk.

The project therefore addresses the following overarching question:

How can data analytics, business intelligence, external data enrichment, and machine learning be combined to analyze e-commerce performance and identify opportunities for more informed business decisions?

1.3 Project Objectives and Research Questions

The main objective of this project is to conduct a comprehensive analysis of the Olist Brazilian E-Commerce Public Dataset and translate the resulting findings into a structured business intelligence report.

The specific objectives are to:

- Prepare and assess the quality of the source data before conducting business analysis.
- Analyze order volume, revenue evolution, and overall sales performance.
- Examine product and category performance using revenue and sales-volume indicators.
- Investigate customer purchasing frequency, spending behavior, and RFM-based segments.
- Analyze seller contribution and the concentration of revenue across sellers.
- Evaluate delivery performance, including late-delivery rates and geographic variation.
- Enrich a subset of order records with historical weather observations using the Open-Meteo API.
- Explore the association between observed weather conditions and delivery delays.
- Develop and evaluate classification models for identifying orders with elevated late-delivery risk.
- Consolidate selected indicators and findings into interactive Power BI dashboards.
- Formulate evidence-based business recommendations while acknowledging analytical limitations.

These objectives are addressed through the following research questions:

1. How did sales and order activity evolve over the observed period?
2. Which product categories contributed most to delivered revenue and sales volume?
3. What patterns characterize customer purchasing frequency, spending, and RFM segments?
4. How is seller revenue distributed, and how concentrated is it among sellers?
5. How did delivery performance vary across geographic areas and over time?
6. What association, if any, was observed between historical weather conditions and delivery delays in the matched sample?
7. How effectively can classification models identify late-delivery orders, and what trade-offs arise when selecting a prediction threshold?
8. How can the findings be presented in a business intelligence dashboard to support further analysis?

1.4 Scope of the Study

This study uses the historical Olist Brazilian E-Commerce Public Dataset. The analysis focuses on the available order, customer, product, seller, and order-item records, with additional weather observations incorporated for a subset of orders.

The commercial analysis covers order activity, product revenue, category performance, customer purchasing behavior, and seller revenue contribution. The operational analysis focuses on delivery outcomes, including delivery duration, estimated-versus-actual delivery performance, and geographic variation.

Customer segmentation is conducted through an RFM framework based on recency, frequency, and monetary value. The predictive modeling component focuses on classifying late-delivery outcomes using selected order and geographic features. The weather component is exploratory and examines observed associations within the subset for which weather data could be matched.

The project's reporting deliverable is an interactive Power BI dashboard covering sales, products and categories, orders and delivery, customers, and sellers.

The study is limited to the information available in the dataset and the constructed analytical samples. It does not provide a complete assessment of business profitability because comprehensive cost data are not available. The historical findings should not be assumed to represent current Olist operations, and observed associations should not be interpreted as causal relationships without further evidence.

1.5 Report Structure

This report is organized into nine main chapters.

Chapter 1 introduces the project context, problem statement, objectives, research questions, and scope. Chapter 2 describes the dataset, data structure, preparation process, analytical workflow, and key metric definitions.

Chapter 3 presents the exploratory analysis of business performance, including sales evolution, product categories, geographic activity, and seller contribution. Chapter 4 focuses on customer analytics, purchasing frequency, spending behavior, and RFM segmentation.

Chapter 5 examines delivery performance and the exploratory integration of historical weather data. Chapter 6 presents the late-delivery prediction task, feature preparation, model comparison, evaluation results, and operational interpretation.

Chapter 7 describes the Power BI data model and dashboard pages. Chapter 8 integrates the principal findings and develops business recommendations. Finally, Chapter 9 summarizes the project's contributions, discusses limitations, and identifies potential directions for future work.

2. Dataset and Analytical Framework

2.1 Olist Dataset Overview

The project uses the Olist Brazilian E-Commerce Public Dataset, a historical dataset made available through Kaggle. It contains records related to e-commerce orders, customers, products, sellers, and order items. These datasets can be connected through identifiers such as order IDs, customer IDs, product IDs, and seller IDs, allowing the analysis to combine commercial and operational information.

The dataset was selected because its relational structure supports analysis across several dimensions of e-commerce activity. Order and order-item records provide the basis for measuring sales and delivery outcomes, while customer and seller information enables geographic and behavioral analysis. Product records and category translations support category-level comparisons.

The dataset contains **99,441 order records** and **112,650 order-item records**. The customer dataset also contains 99,441 records, while the product and seller datasets contain 32,951 and 3,095 records, respectively. These record counts describe the source tables and should not be interpreted as counts of unique customers, delivered orders, or active sellers without applying the relevant identifiers and filters.

The analysis uses the dataset as a historical snapshot. Therefore, the results describe the records and period available in the source data rather than current e-commerce activity.

Source: Olist Brazilian E-Commerce Public Dataset, available on Kaggle.

Table	Records	Analytical purpose
Orders	99,441	Order status, purchase and delivery dates
Customers	99,441	Customer identifiers and geographic information
Order Items	112,650	Product prices, freight, products, and sellers per order

Products	32,951	Product attributes and category
Sellers	3,095	Seller identifiers and geographic information
Category Translation	71	Portuguese-to-English category names

Table 1. Overview of the principal Olist dataset tables.

Note: Record counts refer to source tables before analytical filtering.

2.2 Data Sources and Database Structure

The project combines the Olist transactional dataset with historical weather observations retrieved from the Open-Meteo API.

The Olist data were organized in a MySQL database named `olist_analytics`. The relational structure connects orders to customers and order items, and connects order items to products and sellers. Product categories were linked to the category-translation table to support readable category-level reporting.

The principal relationships used in the Power BI model were:

- Customers to Orders through `customer_id`.
- Orders to Order Items through `order_id`.
- Products to Order Items through `product_id`.
- Sellers to Order Items through `seller_id`.
- Category Translation to Products through `product_category_name`.

These relationships enabled measures and visualizations to combine information from multiple tables while preserving the relevant transaction-level context.

Historical weather data were obtained separately and matched to a subset of orders using geographic and date information. This enrichment was used for exploratory analysis of weather conditions and delivery outcomes; it was not treated as a complete weather record for every order.

The delivery-risk prediction output was also maintained as a separate dataset in Power BI. It was used to examine model results without automatically joining predictions into the transactional model.



Figure 1. Relational data model used for the Olist analysis.

2.3 Data Preparation and Quality Assessment

Data preparation was conducted before the business analysis to improve consistency across the source tables and ensure that the resulting metrics were calculated from clearly defined analytical populations.

The preparation process included inspecting table dimensions, checking identifiers and relationships, preparing date fields for time-based analysis, and creating derived variables required for sales, customer, and delivery analysis. The work was conducted primarily in Python, with the prepared relational data subsequently queried in MySQL.

The source tables were checked for referential consistency. The integrity checks performed for the principal order, item, and product relationships did not identify orphan records in the imported tables.

Several analytical datasets were then constructed for specific purposes. These included delivered order-item records for delivered-revenue and product analysis, customer-level features for purchasing behavior and RFM segmentation, a weather-enriched order sample for exploratory comparison, and a modeling dataset restricted to orders with the required delivery information.

Because these analyses use different filters and populations, their record counts and KPI values are not always directly interchangeable. The report therefore identifies the relevant population alongside metrics where this distinction affects interpretation.

Analytical population	**Record count**
All source orders	99,441
Delivered orders	96,478
Delivered order items	110,197
Unique customer identifiers in source data	96,096
Customers in final RFM dataset	93,358
Orders in final weather-matched sample	40,621

Delivered orders with both delivery dates available for modeling	96,476

Table 2. Analytical populations used in the project

Note: These populations serve different analytical purposes. Counts should not be summed or compared as if they represented the same unit of analysis.

2.4 Analytical Workflow and Tools

The project was organized as a sequential analytical workflow, with each stage contributing to the development of the final business intelligence deliverable. The workflow combined data preparation, relational querying, exploratory analysis, external data enrichment, predictive modeling, and dashboard development.

2.4.1 Data Preparation and Exploratory Analysis with Python

Python was used to inspect and prepare the source data, construct analytical datasets, and conduct exploratory analysis. The work included examining table dimensions and data quality, preparing date variables, combining order and order-item information, and deriving customer-level metrics.

Pandas supported data manipulation and aggregation. The prepared datasets were used to investigate sales performance, delivery outcomes, product categories, customer purchasing patterns, and RFM-based customer segments.

2.4.2 Relational Analysis with MySQL

MySQL was used to organize the source tables within the `olist_analytics` database and conduct structured business analysis. SQL queries were developed to calculate key performance indicators and examine performance across time, customer types, product categories, sellers, and geographic areas.

The SQL work included aggregations, common table expressions, ranking, and customer-level calculations. This stage provided a relational approach to validating and extending findings from the Python analysis.

2.4.3 External Data Enrichment through the Open-Meteo API

The project incorporated historical weather observations retrieved through the Open-Meteo API. Geographic information was used to identify locations for weather retrieval, and the resulting observations were matched to a subset of orders using location and date.

This stage enabled an exploratory comparison of delivery outcomes under different observed weather conditions. Since the matched dataset covered only part of the order population, the findings were interpreted within the limits of that sample.

2.4.4 Predictive Modeling with Machine Learning

The modeling stage focused on classifying late-delivery outcomes. Logistic Regression and Random Forest were developed and evaluated using a dataset of delivered orders with the required delivery-date information.

Model performance was examined using classification metrics, including ROC-AUC, precision, recall, and F1-score. Different probability thresholds were also investigated to understand the trade-off between identifying late orders and generating false alerts.

The resulting predictions were treated as potential signals for operational review rather than definitive statements about future delivery outcomes.

2.4.5 Business Intelligence and Visualization with Power BI

Power BI was used to consolidate selected business metrics and analytical findings into an interactive dashboard. The report was organized into dedicated pages covering sales, product and category intelligence, orders and delivery, customer overview, and seller overview.

The dashboard provides a business-facing layer over the underlying analytical work, allowing users to explore selected performance indicators and comparisons through a common reporting interface.

2.4.6 Integration of Analytical Results

The final stage connects the outputs of the different tools into a unified interpretation. Python and SQL provide the underlying analysis, the API enrichment adds an external exploratory dimension, machine learning evaluates the potential for delivery-risk prioritization, and Power BI presents selected results for business exploration.

This integrated workflow supports the transition from raw transactional records to descriptive findings, exploratory analysis, predictive results, and business recommendations.

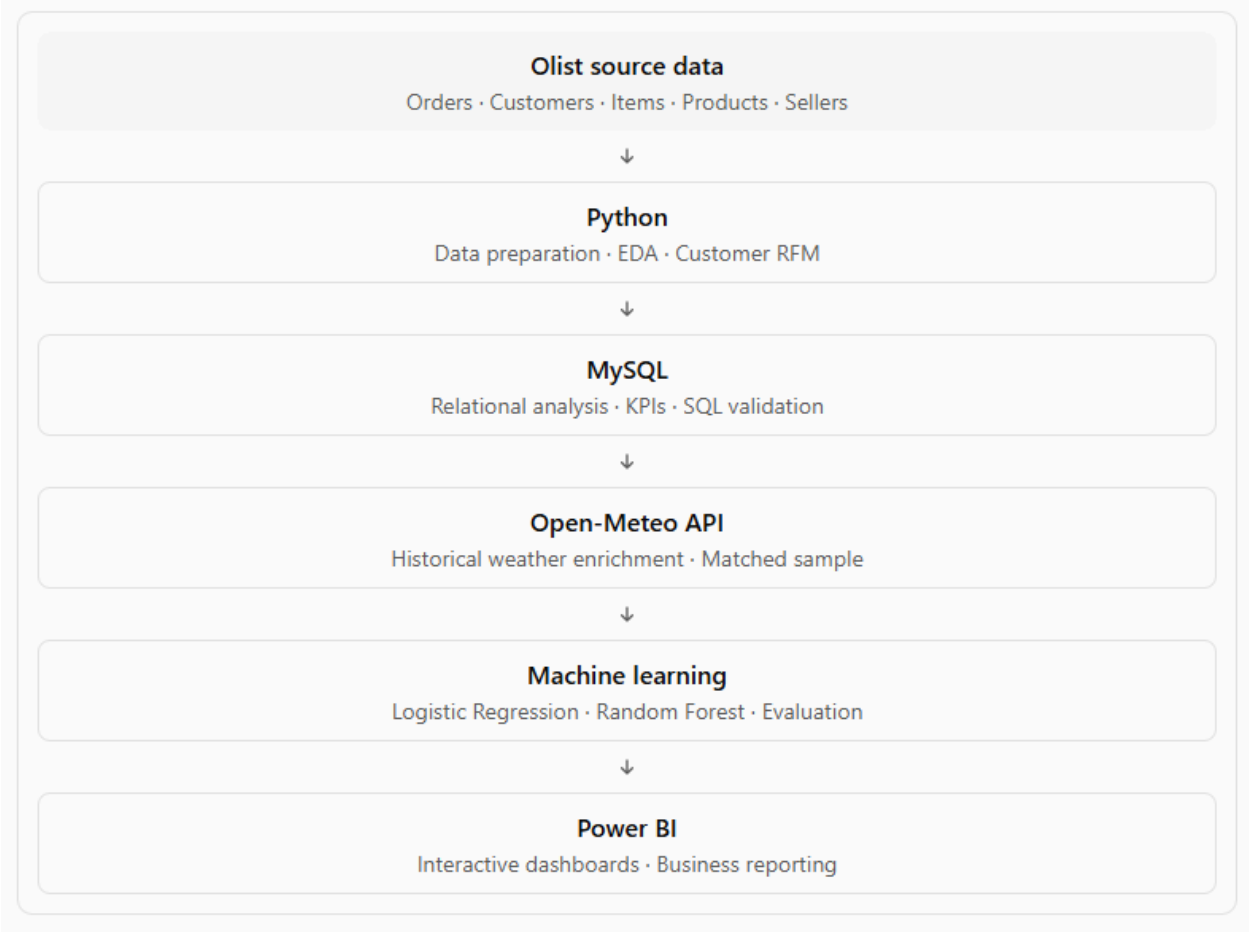


Figure 2. End-to-end analytical workflow of the Olist project.

2.5 KPI Definitions and Measurement Scope

Key performance indicators (KPIs) were used to summarize the commercial and operational characteristics of the dataset. To support consistent interpretation, each KPI must be associated with a clearly defined calculation and analytical population.

2.5.1 Sales and Order KPIs

Total Orders represents the distinct number of order IDs in the order dataset, without restricting the calculation to delivered orders.

Delivered Orders represents the distinct number of orders whose recorded order status is `delivered`.

Completion Rate is calculated as the number of delivered orders divided by the total number of orders in the relevant dataset context.

Product Revenue is calculated as the sum of order-item prices across all order statuses. It is therefore not restricted to delivered orders.

Delivered Product Revenue is calculated as the sum of order-item prices associated with orders whose status is `delivered`.

Freight Revenue represents the sum of freight values recorded in the order-item dataset. The relevant order-status scope must be stated when this measure is used.

Average Order Value (AOV) in the Executive Overview is calculated as delivered product revenue divided by delivered orders. This definition excludes freight and uses delivered orders as its denominator.

2.5.2 Product and Category KPIs

Product and category performance is evaluated using delivered product revenue, delivered units sold, and average product price.

Delivered product revenue measures the sum of item prices for delivered orders. Delivered units sold counts order-item records associated with delivered orders. Average product price represents the average item price within the relevant filter context.

These measures describe sales value and volume; they do not represent product profitability because comprehensive product-cost data are not included.

2.5.3 Customer KPIs

Customer analysis distinguishes customer records, unique customer identifiers, and the customer population used in the RFM analysis.

The repeat-customer rate is calculated using the customer population in the relevant customer analysis and identifies customers whose recorded purchase frequency is greater than one. The denominator must be interpreted according to the population used in that analysis.

Average customer spend is a customer-level measure and must be distinguished from average order value. In this project, the Power BI Customer Overview reports an average customer spend of approximately €141.62. Its interpretation depends on the revenue measure and customer denominator applied in the visual.

2.5.4 Delivery KPIs

Delivery performance is evaluated using delivery duration, late-order count, and late-delivery rate.

Average delivery days measures the elapsed time between purchase and customer delivery for orders with a recorded delivery date, subject to the applicable report filters.

Late Orders — Both Dates Available counts delivered orders for which both the actual and estimated delivery dates are available and the actual delivery date is later than the estimated date.

Late Rate is calculated as late orders divided by delivered orders with both relevant dates available. This denominator is used to avoid treating orders without the required date information as confirmed on-time or late deliveries.

2.5.5 Predictive Modeling Metrics

The predictive modeling stage evaluates classification performance using ROC-AUC, accuracy, precision, recall, and F1-score.

These metrics describe different aspects of model performance. In particular, precision indicates the proportion of orders flagged as late that were actually late, while recall indicates the proportion of actual late orders identified by the model.

The selected classification threshold determines the balance between identifying late orders and generating false alerts. The reported threshold is therefore an analytical operating point, not a universal business rule.

2.5.6 Interpretation Across Different Analytical Populations

The project includes several datasets constructed for different analytical purposes. As a result, a KPI calculated from all source orders may differ from a metric calculated only for delivered orders, customers included in the RFM dataset, or orders with complete delivery information.

Throughout the report, the scope of each KPI should be considered alongside its value. Comparisons should only be made when the underlying populations, filters, and calculation methods are compatible.

KPI	Definition	Scope
Total Orders	Distinct count of order IDs	All orders
Delivered Orders	Distinct count of orders with status `delivered`	Delivered orders

Completion Rate	Delivered orders ÷ total orders	Same filter context
Product Revenue	Sum of item prices	All order statuses
Delivered Product Revenue	Sum of item prices	Delivered orders
Average Order Value	Delivered product revenue ÷ delivered orders	Delivered orders; excludes freight
Delivered Units Sold	Count of order-item records	Delivered orders
Late Orders	Actual delivery date later than estimated date	Delivered orders with both dates
Late Rate	Late orders ÷ delivered orders with both dates	Complete-date delivery subset
Repeat Customer Rate	Repeat customers ÷ customers in the defined customer population	Customer analysis population

Table 3. Core KPI definitions and scope

3.1 Overall Sales and Order Performance

The first stage of the business performance analysis examined the overall volume of orders, delivery completion, and product revenue recorded in the Olist dataset. These indicators establish the scale of the analyzed e-commerce activity and provide a baseline for the more detailed product, customer, seller, and delivery analyses that follow.

The source dataset contains **99,441 orders**. Of these, **96,478 orders** were recorded with a `delivered` status, corresponding to a completion rate of approximately **97.02%** when calculated against all orders.

The revenue analysis distinguishes between product revenue across all order statuses and product revenue associated with delivered orders. The sum of order-item prices across all statuses amounted to approximately **€13,591,643.70**. When restricted to delivered orders, product revenue amounted to approximately **€13,221,498.11**.

This distinction is important because the two measures represent different order populations. The all-status figure provides a view of the product value recorded across the dataset, while delivered product revenue focuses on orders recorded as delivered. Neither measure should be interpreted as net profit, since the available analysis does not account for all business costs.

The results provide a baseline for the subsequent analysis of revenue evolution, category contribution, customer spending, seller concentration, and delivery performance. In particular, the difference between total order volume and delivered order volume establishes the importance of examining order outcomes alongside commercial activity.

Overall, the initial analysis indicates that the majority of recorded orders reached delivered status, while the remaining orders require separate consideration according to their recorded status and the purpose of each subsequent analysis.

Indicator	Value
Total orders	99,441
Delivered orders	96,478
Completion rate	97.02%
Product revenue — all order statuses	€13,591,643.70

Delivered product revenue	€13,221,498.11

Table 4. Overall sales and order performance

Note: Product revenue is calculated from order-item prices and excludes freight. The all-status and delivered revenue figures use different populations.

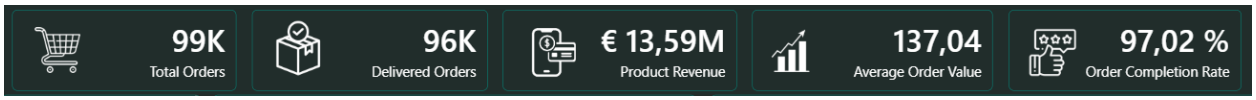


Figure 3. Sales Overview dashboard and headline performance indicators.

3.2 Revenue Evolution Over Time

The temporal analysis examined how product revenue evolved throughout the period covered by the Olist dataset. Monthly revenue was used to observe changes in commercial activity over time, while the SQL analysis provided a structured way to calculate monthly totals and compare revenue across consecutive periods.

The analysis showed an overall increase in monthly revenue during the observed period. The highest monthly revenue identified in the analysis occurred in **November 2017**, at approximately **€988,000**.

This pattern provides evidence of increasing recorded sales activity during the period examined. However, the available time series does not establish a strong recurring seasonal pattern. The observed increase should therefore not be interpreted as proof of sustained future growth or attributed to a specific business cause without additional evidence.

Monthly revenue is also affected by the number and value of orders recorded in each period. Consequently, changes in revenue should be considered alongside order volume and other business indicators when interpreting performance.

The temporal analysis provides a foundation for further investigation into the drivers of revenue variation, including product-category contribution, customer purchasing behavior, and geographic activity.

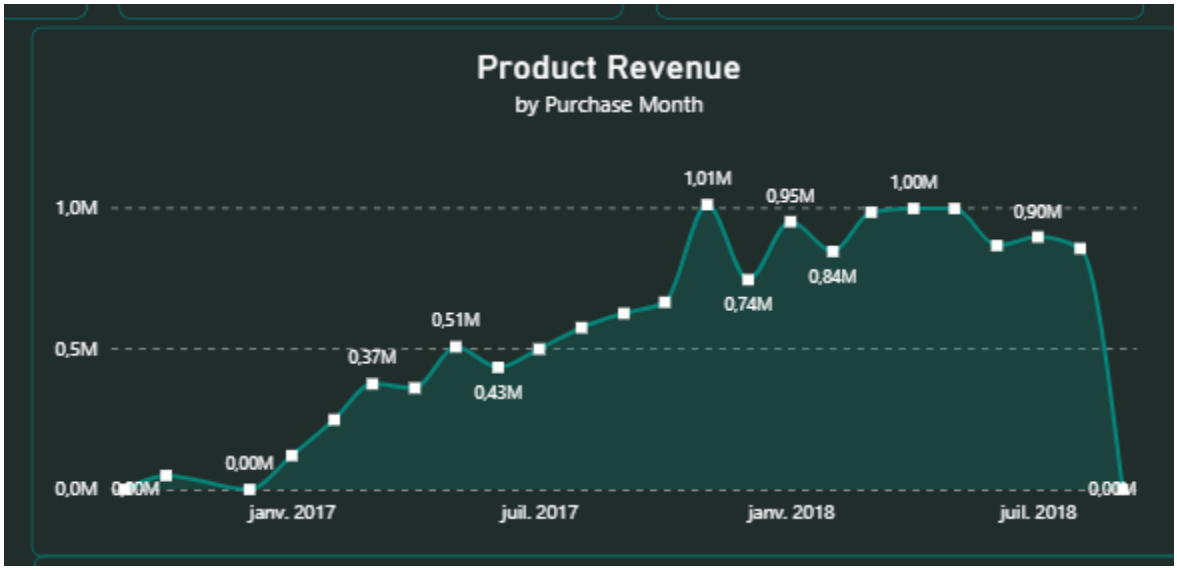


Figure 4. Monthly product revenue over the observed period.

Monthly product revenue increased over the observed period, reaching its highest point in November 2017 at approximately €1.01 million. Revenue remained relatively high during several months of 2018, although the final month shown records a sharp drop that may reflect an incomplete observation period.

3.3 Product and Category Performance

Product and category analysis provides insight into the types of products generating revenue and the categories contributing most to the platform's sales. This analysis focuses on delivered orders to reflect completed transactions and their associated product revenue.

3.3.1 Revenue Contribution by Product Category

The analysis identified differences in revenue contribution across product categories. Health and beauty generated approximately €1.23 million in delivered product revenue, followed by watches and gifts at approximately €1.17 million and bed, bath and table at approximately €1.02 million.

Sports and leisure and computers and accessories also represented substantial revenue-generating categories, with approximately €955,000 and €889,000 respectively.

These results indicate that revenue is distributed across several product categories rather than being exclusively concentrated in a single category. The categories generating the highest revenue may therefore provide useful starting points for examining product demand, assortment decisions, and commercial performance.

3.3.2 Sales Volume and Product Price

Revenue alone does not fully describe category performance. Units sold provide a complementary measure of customer demand, while average product price helps distinguish categories driven by sales volume from those generating revenue through higher-priced products.

Bed, bath and table recorded the highest unit volume among the leading revenue categories, with approximately 10,953 units sold. This illustrates why revenue and sales volume should be examined together: a category's position in terms of revenue does not necessarily correspond to its position in terms of units sold.

The analysis therefore considers both revenue contribution and sales volume to provide a broader view of product performance.

3.3.3 Implications for Business Analysis

The category results can support several areas of further investigation:

- **Assortment planning:** Identify categories that contribute substantially to delivered revenue or sales volume.
- **Demand analysis:** Compare units sold with revenue to distinguish high-volume categories from categories with higher product prices.
- **Performance monitoring:** Track category-level revenue and volume over time to identify changes in demand.
- **Further investigation:** Examine individual products within leading categories to determine whether category performance is broadly distributed or driven by a smaller number of products.

These findings describe historical performance in the available dataset. They do not, on their own, establish the causes of category-level demand or guarantee future sales performance.

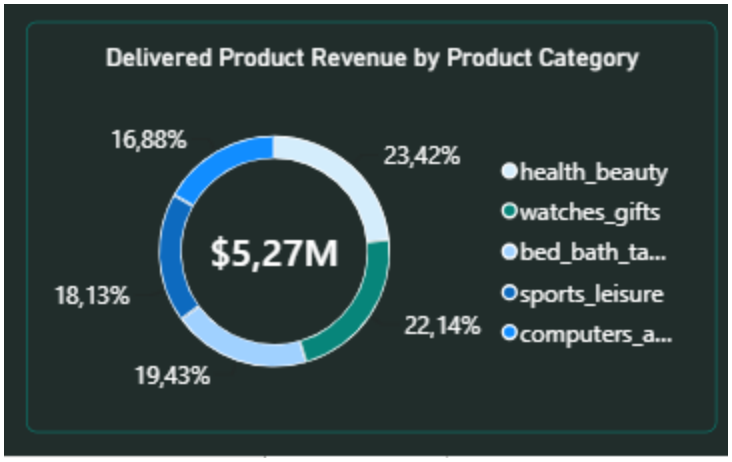


Figure 5. Delivered product revenue by category

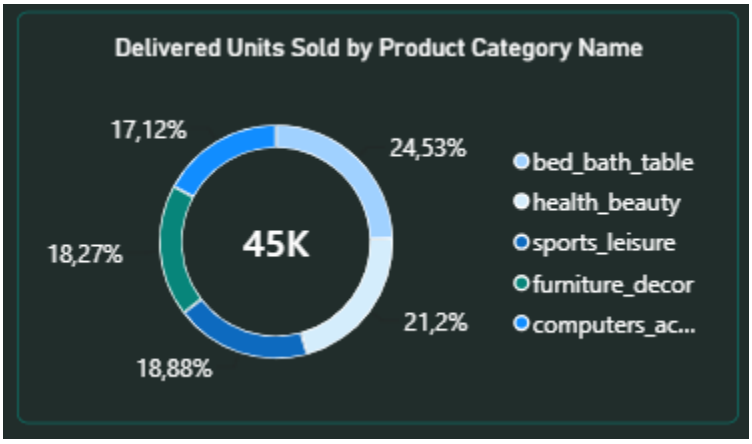


Figure 6. Units sold by product category

3.4 Geographic and Seller Performance

Geographic and seller-level analysis adds an operational perspective to the sales results by examining where customer demand is concentrated and how seller activity contributes to platform revenue. This section considers customer-state order activity and seller revenue, using the available order, customer, seller, and order-item data.

3.4.1 Geographic Distribution of Customer Demand

Customer-state analysis reveals differences in the geographic distribution of delivered orders. The distribution is useful for understanding where the platform's completed transactions are concentrated and for identifying regions that may warrant more detailed operational or commercial analysis.

However, order volume should not be interpreted as a direct measure of market potential. The dataset describes observed platform activity, not the full population of potential customers in each state. Differences in population, platform penetration, and other market conditions are not independently measured in this analysis.

3.4.2 Seller Revenue and Geographic Activity

Seller-level analysis provides another view of revenue generation. The SQL analysis identified the following seller locations among the highest-revenue cities:

- Guariba, São Paulo (SP): approximately €226,988
- Lauro de Freitas, Bahia (BA): approximately €217,940
- Ibitinga, São Paulo (SP): approximately €196,882
- Sumaré, São Paulo (SP): approximately €190,917
- Itaquaquecetuba, São Paulo (SP): approximately €186,570

These figures represent revenue associated with seller locations in the analyzed data. They should not be interpreted as revenue generated by every seller in a city, nor as evidence that location alone explains seller performance.

3.4.3 Seller Revenue Concentration

The seller revenue analysis also examined how revenue is distributed across sellers. The cumulative contribution of the highest-revenue sellers was:

Share of sellers considered	**Cumulative revenue contribution**
Top 10%	41.10%
Top 20%	56.62%
Top 30%	67.68%
Top 40%	76.35%

Top 50%	83.31%

Table 5. Seller revenue analysis

The results show that seller revenue is not evenly distributed: the highest-revenue 10% of sellers account for 41.10% of the revenue included in this analysis, while the highest-revenue 50% account for 83.31%.

This concentration is relevant for performance monitoring and dependency analysis. It also means that aggregate platform results may be influenced substantially by a subset of sellers. Revenue concentration alone, however, does not establish that the platform is operationally dependent on those sellers; that assessment would also require information about product substitutability, seller availability, and the consequences of seller disruption.

3.4.4 Business Implications

The geographic and seller findings can inform further investigation in three areas:

- **Regional monitoring:** Compare delivered order volumes across customer states and investigate differences in activity.
- **Seller performance:** Monitor seller-level revenue and order activity rather than relying only on city-level aggregates.
- **Concentration risk:** Track the revenue share of leading sellers over time and investigate whether high concentration is accompanied by limited alternatives or operational vulnerabilities.

Overall, geographic and seller analysis complements the preceding sales and category results by showing how completed transactions are distributed across locations and sellers. The findings describe observed historical activity and should be interpreted within the scope and limitations of the dataset.



Figure 7. Delivered orders by customer state



Figure 8. Seller revenue concentration



Figure 9. Seller geographic distribution

4. Customer Analytics and Segmentation

4.1 Customer Base and Purchase Frequency

Customer analytics provides a perspective on the platform's performance beyond individual transactions. By examining how often customers purchase and how spending is distributed across the customer base, it becomes possible to distinguish between customer acquisition, repeat purchasing, and customer value.

4.1.1 Overview of the Customer Base

The original dataset contains 99,441 customer records and 96,096 distinct customer identifiers based on `customer_unique_id`. These identifiers allow purchases associated with the same customer to be analyzed across multiple orders.

For the RFM analysis, the final customer-level dataset contains 93,358 customers. This population differs from the distinct-customer count in the original dataset because it reflects the records retained for the RFM analysis.

4.1.2 Purchase Frequency and Repeat Customers

The purchase-frequency analysis shows that most customers in the analyzed population placed only one order. In the Python analysis, 93,099 customers were classified as one-time customers, representing 96.88% of the analyzed customer base. The remaining 2,997 customers, or 3.12%, were classified as repeat customers.

The SQL analysis also identified a low repeat-purchase share, although its customer counts differ from the Python results because the analyses use different underlying populations and processing choices. The SQL results show 90,557 one-time customers and 2,801 repeat customers, with repeat customers representing approximately 3% of the analyzed group.

These results indicate that repeat purchasing represents a relatively small share of observed customer activity. They do not, by themselves, establish why customers did not return. The dataset does not directly measure customer satisfaction, reasons for churn, or whether customers continued purchasing through other channels.

4.1.3 Purchase Frequency and Customer Value

Purchase frequency is useful when interpreted alongside spending. A customer who places several orders may generate a different pattern of value from a customer who makes a single high-value purchase.

The SQL analysis found that repeat customers had an average order value of approximately €260.05, compared with approximately €137.96 for one-time customers. Repeat customers contributed approximately 5.51% of the revenue in that SQL analysis, while one-time customers contributed approximately 94.49%.

This comparison highlights the importance of distinguishing between order value, total customer spending, and purchase frequency. A higher average order value among repeat customers does not necessarily mean that repeat customers account for most platform revenue, particularly when they represent a small share of the customer base.

4.1.4 Interpretation

The customer-base analysis establishes the context for the RFM segmentation in the following sections. The low observed repeat-purchase share makes customer retention an important area for further investigation, while the spending differences suggest that frequency and monetary value should be considered together.

These findings are descriptive. They identify patterns in the available transaction data but do not establish the causes of repeat purchasing or predict whether an individual customer will return.

4.2 Customer Spending and Monetary Value

While purchase frequency describes how often customers place orders, monetary analysis examines the value associated with their purchases. Considering both dimensions helps distinguish customers who purchase frequently from those whose individual or cumulative spending is comparatively high.

4.2.1 Average Order Value

The SQL analysis found an average order value of approximately €137.96 for one-time customers and €260.05 for repeat customers. This difference indicates that repeat customers placed orders with a higher average value in the analyzed population.

Average order value should not be confused with total customer spending. It measures the average value per order, whereas total customer spending aggregates purchases associated with a customer over the analysis period.

4.2.2 Customer Spending Distribution

The RFM analysis provides a complementary view by comparing customer segments according to their monetary value. Average spending varied across the segments:

RFM segment	Approximate average customer spend
Champions	€865
Loyal Customers	€649
At Risk	€433
Other	€202
Potential Loyalists	€169
Hibernating / Low Value	€56

Table 6. RFM Segmentation and Customer Characteristics

The figures show substantial variation in monetary value across the RFM segments. Champions and Loyal Customers have higher average spending in this analysis, while Hibernating / Low Value customers have a lower average.

Importantly, the “Other” segment should not automatically be interpreted as a loyal customer group. It includes a broad set of customers, including one-time purchasers, and its monetary value does not by itself establish a sustained relationship with the platform.



Figure10. Average Customer Spend by Segment

4.2.3 Revenue Concentration Among Customers

The customer-level revenue analysis also examined the contribution of high-spending customers. The top 10% of customers accounted for approximately 38.25% of revenue, while the top 20% accounted for approximately 53.52%.

This indicates that a substantial share of revenue was generated by a relatively small portion of the customer base. Such concentration makes it useful to monitor both the spending patterns of high-value customers and the broader distribution of customer value.

Revenue concentration should be interpreted alongside purchase frequency and retention. A customer may contribute substantial revenue through a single purchase without becoming a repeat customer.

4.2.4 Interpretation

The spending analysis shows why customer value cannot be described using a single metric. Average order value, cumulative spending, and purchase frequency provide different perspectives on customer behavior.

Taken together, the results suggest that customer analytics should monitor both high-value purchasing and the development of repeat relationships. The available transaction data

identifies spending patterns, but it does not directly explain customer motivations or establish the reasons behind differences in monetary value.

4.3 RFM Methodology and Customer Segments

To provide a structured view of customer behavior, the project applied RFM analysis. RFM stands for Recency, Frequency, and Monetary value. It combines three dimensions of purchasing activity to distinguish customer groups according to how recently they purchased, how often they purchased, and how much they spent.

4.3.1 RFM Dimensions

The analysis used the following measures:

- **Recency (R):** The number of days since a customer's most recent purchase. A lower recency value indicates a more recent purchase.
- **Frequency (F):** The number of orders associated with the customer during the analysis period. Higher frequency indicates more repeated purchasing.
- **Monetary value (M):** The customer's accumulated product spending, used to represent the monetary value of their purchases.

The final RFM dataset contained 93,358 customer records and included the calculated R, F, and M measures, their scores, and the resulting RFM score.

4.3.2 Scoring Approach

The project converted the three RFM dimensions into scores to support customer grouping. Frequency scoring used the following thresholds:

Number of orders	Frequency score
1	1
2	2

3	3
4–5	4
6 or more	5

Table 7. Frequency of Number of Orders

The RFM scores were then used to assign customers to segments. This approach provides a simplified representation of customer behavior and makes it possible to compare groups using a consistent segmentation framework.

4.3.3 Customer Segment Distribution

The resulting segmentation included six groups: Other, Potential Loyalists, Hibernating / Low Value, At Risk, Champions, and Loyal Customers.

The largest groups were Other, representing 43.36% of customers, and Potential Loyalists, representing 40.06%. Hibernating / Low Value customers accounted for 16.47% of the RFM population. At Risk, Champions, and Loyal Customers each represented less than 0.1% of customers.

These results show that the RFM population is concentrated in the three largest segments, while the remaining segments contain comparatively small proportions of customers.

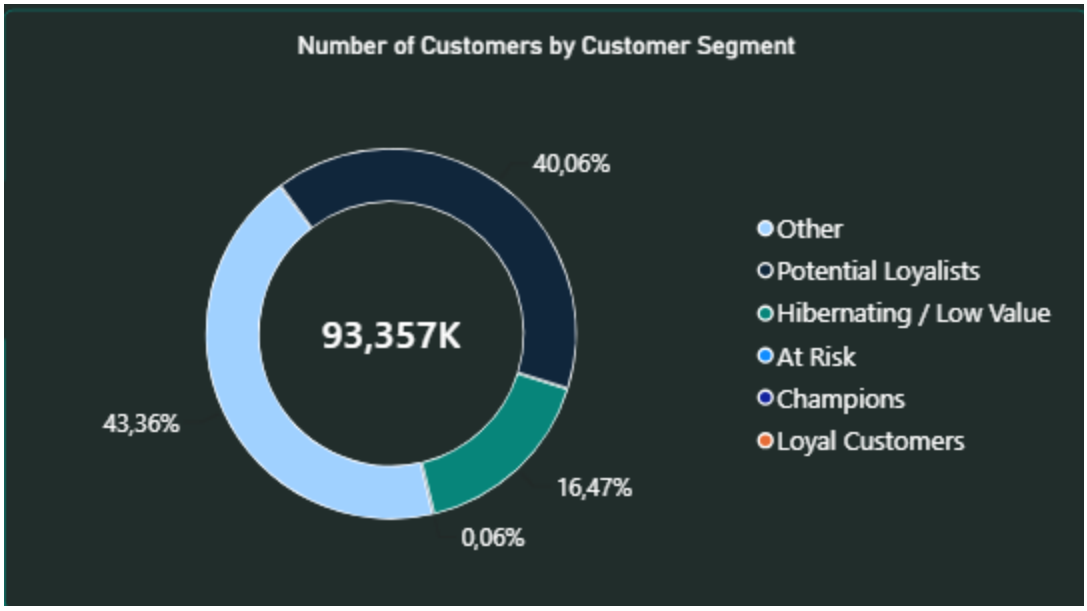


Figure 11. Customer Segment distribution

4.3.4 Interpretation and Limitations

The segment labels provide a practical way to summarize differences in purchasing behavior, but they should be interpreted in relation to the scoring rules and the observed transaction history.

In particular, the large Other segment should not be treated as synonymous with loyal customers. A high monetary value can result from a high-value purchase without repeated purchasing. Similarly, a segment label is a descriptive classification, not a direct measurement of customer satisfaction or future behavior.

The RFM results therefore provide a basis for describing customer groups and identifying areas for further analysis. They do not establish the reasons behind customer behavior or guarantee that customers will follow a particular future purchasing pattern.

4.4 Segment Revenue Contribution and Retention Implications

Customer segmentation becomes more informative when the distribution of customers is considered alongside each segment's contribution to revenue. This comparison helps

distinguish segments that account for a large share of customers from those that generate a disproportionate share of spending.

4.4.1 Revenue Contribution by Segment

The RFM analysis shows that the **Other** segment accounted for 53.09% of revenue, while **Potential Loyalists** contributed 40.95%. **Hibernating / Low Value** customers represented 5.56% of revenue. The remaining segments—At Risk, Champions, and Loyal Customers—each contributed less than 0.2%.

These results indicate that most revenue in the RFM analysis was associated with the three largest segments. The small customer shares of Champions and Loyal Customers mean that their individual average spending should be interpreted alongside their limited contribution to total revenue.

4.4.2 Comparing Customer Share and Revenue Share

The Other segment represented 43.36% of customers and 53.09% of revenue. Potential Loyalists represented 40.06% of customers and 40.95% of revenue. Hibernating / Low Value customers accounted for 16.47% of customers but 5.56% of revenue.

This comparison highlights differences in the distribution of customer numbers and monetary contribution. In particular, the Hibernating / Low Value segment represents a larger share of customers than of revenue, while the Other segment contributes a larger share of revenue than its share of customers.

These differences are descriptive and should be interpreted in the context of the RFM scoring and segment definitions.

4.4.3 Retention Implications

The segmentation provides a framework for organizing further customer-retention analysis:

- **Potential Loyalists:** Examine whether these customers develop repeat-purchase behavior over time.
- **Hibernating / Low Value:** Monitor recency and subsequent purchasing activity to understand whether customers return after a period of inactivity.
- **Champions and Loyal Customers:** Track their purchasing patterns while recognizing that these groups represent very small portions of the analyzed customer base.
- **Other:** Avoid assuming that all customers in this broad segment have the same relationship with the platform; examine frequency and monetary value separately where possible.

These are analytical directions rather than tested intervention outcomes. The transaction data does not show whether a specific retention campaign would change customer behavior.

4.4.4 Summary

The segment-level comparison demonstrates that customer count, purchase frequency, and revenue contribution provide complementary perspectives. The largest segments account for most of the observed revenue, while the smaller high-value segments represent a limited share of customers.

RFM segmentation therefore supports a more differentiated understanding of the customer base. It can help structure monitoring and further investigation, but the results should not be interpreted as evidence that particular retention actions will necessarily increase revenue.

5. Delivery and Weather Analysis

5.1 Delivery Performance Overview

Delivery performance is an important operational dimension of e-commerce because it reflects the time between purchase and customer receipt. This section examines delivery duration and the distribution of delivery outcomes in the Olist dataset.

5.1.1 Delivery Time

For orders with an available customer delivery date, delivery duration was calculated as the number of days between the purchase timestamp and the actual delivery timestamp.

The Python analysis included 96,476 orders with the required delivery information. The average delivery duration was approximately 12.1 days, with a standard deviation of approximately 9.6 days. The observed minimum was 0 days and the maximum was 209 days.

The wide range indicates substantial variation in delivery duration. While the average provides a useful summary, it does not fully represent the experience of every customer, particularly where a small number of orders took considerably longer to arrive.

5.1.2 Delivery Status Distribution

Delivery outcomes were classified by comparing the actual delivery date with the estimated delivery date. Orders delivered before the estimate were classified as early, those delivered on the estimated date as on time, and those delivered after the estimate as late. Orders without a recorded delivery date were classified separately as not delivered.

The initial Python classification produced the following distribution:

Delivery status	Number of orders	Share of classified orders
Early	88,649	89.15%
Late	6,535	6.57%
On Time	1,292	1.30%
Not Delivered	2,965	2.98%

Table 8. Delivery Status Distribution

The results indicate that most orders were classified as early, while late orders represented a smaller share of the classified dataset. The not-delivered category is retained separately because the absence of a recorded delivery date does not establish that an order was delivered.

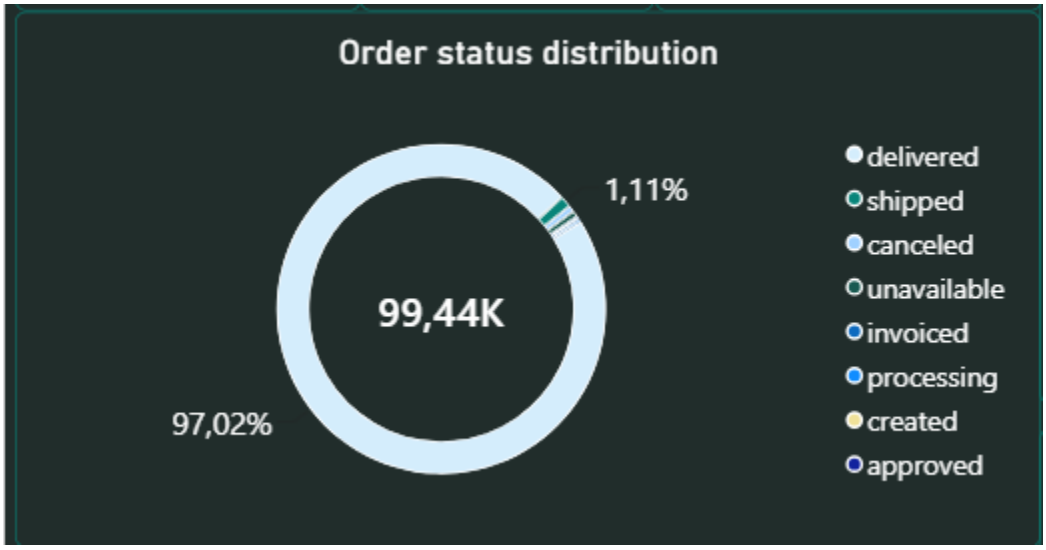


Figure12. Order Status Distribution

5.1.3 Interpreting Delivery Outcomes

The high share of early deliveries suggests that the estimated delivery date was not exceeded for most orders in the analyzed data. However, the late-order share remains relevant for operational monitoring because late delivery can affect a subset of customers even when overall delivery performance appears favorable.

The distinction between early, on-time, late, and not-delivered orders is important. Combining these outcomes into a single average could conceal differences between completed deliveries and orders without a recorded delivery date.

5.1.4 Data Considerations

Delivery metrics depend on the availability and consistency of the underlying timestamps. The analysis identified a small number of records for which a missing customer delivery date affected the initial status classification. Such cases should be treated cautiously rather than automatically assumed to represent a particular delivery outcome.

The delivery-duration and delivery-status measures also use specific populations. Consequently, their denominators may differ, and comparisons should state which orders are included.

Overall, this section establishes the baseline for the more detailed temporal, geographic, and weather-related delivery analysis that follows.

5.2 Late Deliveries Across Time and Geography

Delivery performance can vary across time and geographic areas. Examining these patterns helps identify where late deliveries are more frequently observed and provides context for the subsequent weather analysis.

5.2.1 Monthly Late-Delivery Patterns

The project examined late-delivery rates across months to explore how delivery outcomes changed over the observation period. The late-delivery rate was calculated as the number of late orders divided by the number of delivered orders with both actual and estimated delivery dates available.

The overall late-delivery rate for this population was approximately 6.77%, based on 6,535 late orders among 96,476 eligible delivered orders.

Monthly rates provide a more detailed view than the overall average, but they should be interpreted with attention to the number of orders observed in each month. A high rate in a month with relatively few orders may represent a small number of cases and should not automatically be interpreted as a broad operational deterioration.

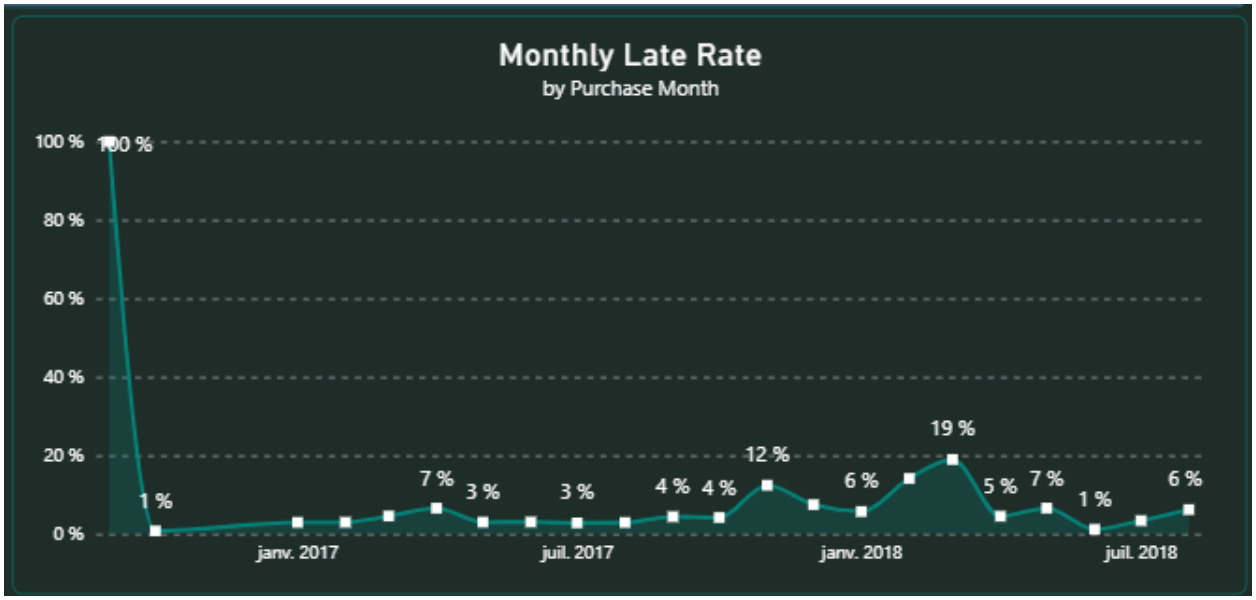


Figure13. Monthly Late Rate

5.2.2 Geographic Differences in Delivery Risk

The analysis also compared delivery outcomes across customer states. The results showed variation in late-delivery rates between states. For example, the observed late-delivery rates were approximately 21.41% in Alagoas (AL), 17.43% in Maranhão (MA), and 15.22% in Sergipe (SE).

These rates should be considered alongside the number of delivered orders in each state. Smaller states may have higher observed rates based on a relatively limited number of transactions, while larger states may account for more late orders in absolute terms despite having lower rates.

The geographic comparison therefore distinguishes between **delivery volume** and **delivery risk**. Both measures are useful, but they answer different questions: volume describes the scale of activity, while the late-delivery rate describes the proportion of eligible deliveries that arrived after the estimate.



Figure 14. Delivery Risk by State

5.2.3 Interpretation

The temporal and geographic analyses demonstrate that an overall delivery metric can conceal differences across individual months and regions. These patterns can help direct further investigation toward areas with elevated observed late-delivery rates.

However, the descriptive comparisons do not establish the causes of these differences. Factors such as shipment distance, seller practices, logistics capacity, and local conditions would require additional data and analysis to assess their respective contributions.

5.3 Weather Data Integration and Analysis

To investigate whether weather conditions were associated with delivery outcomes, the project enriched the e-commerce data with historical weather observations from the Open-Meteo API. This added an external environmental dimension to the delivery analysis.

5.3.1 Weather Data Collection and Integration

Historical weather data was collected for the leading customer cities identified in the dataset. The collection period covered the relevant order-delivery dates, including an extension to October 2016 to account for deliveries recorded during that year.

The resulting weather dataset contained 15,020 city-date observations, covering the period from 11 October 2016 to 31 October 2018. The data was then matched to delivery records using geographic and date information.

The initial merge produced 40,622 eligible orders. One record lacked a corresponding weather observation and was excluded from the weather comparison, leaving a final matched sample of 40,621 orders.

This integration made it possible to compare delivery outcomes across orders associated with different observed weather conditions. The weather-enriched sample is smaller than the full delivery dataset and should therefore be treated as a separate analytical population.

5.3.2 Weather Conditions and Late-Delivery Rates

Within the final matched sample, orders were grouped according to whether rain was recorded. The late-delivery rate was approximately 5.89% for orders associated with no rain and 7.41% for orders associated with rain.

Weather group	Orders	Late orders	Late-delivery rate
No rain	15,355	904	5.89%
Rain	25,266	1,873	7.41%

Table 10. Late-Delivery Rates by Weather Condition

The observed difference was approximately 1.53 percentage points. In this sample, the late-delivery rate was higher among orders associated with rain than among those associated with no rain.

A more detailed comparison by rain intensity showed the following rates:

Weather category	Late-delivery rate
No rain	5.89%
Light rain	7.31%

Heavy rain	7.70%

Table 11. Late-Delivery Rates by Rain Intensity

These results describe an association in the matched dataset. They do not establish that rain caused the observed differences in delivery performance.

5.3.3 Interpretation and Limitations

The weather analysis adds context to the delivery findings by showing that late-delivery rates differed across the weather groups examined. However, several limitations affect the interpretation.

First, the matched weather sample covers only a subset of the full delivery dataset, so the results should not be generalized automatically to all orders. Second, weather conditions may be related to other factors—such as geographic location, shipment distance, and local logistics—that are not isolated by this comparison.

Finally, the weather observations were associated with the actual delivery date. This makes them useful for retrospective analysis, but unsuitable as a direct predictive feature for forecasting a delay before delivery: the actual delivery date is not known at the time such a forecast would be made.

5.3.4 Summary

The weather-enriched analysis identified a higher observed late-delivery rate for orders associated with rain than for those associated with no rain. The difference provides a basis for further investigation into environmental and operational factors affecting delivery outcomes.

The result should be interpreted as an association rather than a causal effect. Further analysis using weather forecasts available at the time of purchase or shipment, together with additional logistics variables, would be required to assess whether weather information could improve prospective delivery-risk monitoring.

5.4 Integrated Interpretation of Delivery and Weather Findings

The delivery analysis identified generally early deliveries alongside a smaller proportion of late orders. Across 96,476 delivered orders with the required date information, 6,535 were classified as late, corresponding to an overall late-delivery rate of approximately 6.77%.

The temporal and geographic analyses showed that delivery outcomes varied across months and customer states. These differences indicate that aggregate delivery metrics should be complemented by more detailed monitoring of specific periods and regions. However, the observed patterns do not establish the operational causes of late delivery.

The weather-enriched analysis added an external dimension to this investigation. In the matched sample of 40,621 orders, the late-delivery rate was approximately 7.41% for orders associated with rain, compared with 5.89% for orders associated with no rain. This difference represents an observed association within the matched sample, not evidence that rain directly caused delays.

Taken together, the findings support monitoring delivery performance through multiple indicators: overall late-delivery rate, delivery duration, monthly variation, geographic differences, and contextual factors such as weather. Each indicator captures a different aspect of delivery performance and should be interpreted with its corresponding population and limitations.

The analysis also establishes an important distinction between retrospective explanation and prospective prediction. Weather recorded for the actual delivery date can help describe conditions associated with completed deliveries, but it is not available in the same form when a delivery-risk prediction must be made in advance. This distinction informs the feature selection and evaluation approach presented in the next chapter.

6. Predictive Modeling

6.1 Modeling Objective and Target Definition

The predictive modeling phase aimed to assess whether order-level information available before delivery could help identify orders at risk of arriving later than their estimated delivery date. This extends the preceding descriptive analysis by moving from understanding historical delivery outcomes toward evaluating a potential early-warning approach.

6.1.1 Modeling Objective

The objective was to develop and evaluate classification models capable of distinguishing late deliveries from deliveries that arrived on or before the estimated date.

The intended business application is to support delivery-risk monitoring by identifying orders that may warrant further review. The model is not intended to confirm that a particular order will be late, nor to replace operational judgment.

6.1.2 Target Definition

The target variable was derived by comparing the actual customer delivery date with the estimated delivery date:

- **Late (1):** Actual delivery date was later than the estimated delivery date.
- **Not late (0):** Actual delivery date was on or before the estimated delivery date.

The modeling dataset included 96,476 delivered orders with the date information required to determine the target. Of these, 6,535 were classified as late and 89,941 as not late.

Late orders represented approximately 6.77% of the modeling population, while approximately 93.23% were not late. This imbalance is important when evaluating model performance because a classifier could achieve high overall accuracy by favoring the majority class while missing many late deliveries.

6.1.3 Modeling Population and Scope

The target was defined using completed deliveries with both actual and estimated delivery dates available. Orders without the required information were excluded from this modeling population.

Consequently, the model evaluates the classification of delivery outcomes within this selected population. Its results should not automatically be generalized to orders that were cancelled, remained undelivered, or lacked the necessary timestamps.

The modeling objective is therefore a retrospective evaluation of delivery-risk classification using the available order data. Prospective deployment would require additional validation to ensure that all input features are available at the intended prediction time and that model performance remains reliable on future orders.

6.2 Feature Selection and Modeling Approach

The models used order-level and geographic information to evaluate whether late deliveries could be identified from characteristics associated with an order. Feature selection focused on variables included in the prepared modeling dataset.

6.2.1 Selected Features

The following features were used in the modeling process:

Feature	Description

total_order_value	Total value associated with the order
total_freight	Freight cost associated with the order
number_of_items	Number of items in the order
number_of_sellers	Number of distinct sellers associated with the order
same_state	Indicates whether the customer and seller are in the same state
purchase_month	Month in which the order was purchased
estimated_delivery_days	Estimated delivery duration
customer_state	Customer's state
seller_state	Seller's state

Table 12. Predictive Modeling Dataset

These variables represent order value, shipment characteristics, timing, and geographic information. Their inclusion allowed the models to consider several dimensions of an order rather than relying on a single predictor.

6.2.2 Train-Test Split

The dataset was divided into training and testing subsets using an 80/20 split. Stratification was applied to preserve the approximate proportion of late and non-late orders in both subsets.

The training subset was used to fit the models, while the test subset was reserved for evaluating their performance on observations not used during model fitting.

6.2.3 Models Evaluated

Two classification approaches were evaluated:

- **Logistic Regression:** A classification model used as a baseline for predicting the probability of a late delivery.
- **Random Forest:** An ensemble model that combines multiple decision trees to capture potentially more complex relationships among the input features.

Evaluating both models provided a comparison between a linear classification approach and a tree-based ensemble approach.

6.2.4 Feature Availability and Interpretation

Although the selected variables are associated with order and delivery characteristics, their availability must be considered in relation to the intended prediction time. In particular, a prospective early-warning system should use only information available when the prediction is made.

The model's results should therefore be understood as an evaluation of the selected feature set, with prospective use requiring additional checks of feature timing, data quality, and performance on future orders.

6.3 Model Evaluation and Performance

The classification models were evaluated on the held-out test set to assess their ability to identify late deliveries. Because late orders represented a relatively small proportion of the modeling population, evaluation considered not only overall accuracy but also the models' ability to detect late cases.

6.3.1 Evaluation Metrics

The following metrics were used:

- **Accuracy:** The proportion of all test observations classified correctly.
- **Precision:** The proportion of orders predicted as late that were actually late.
- **Recall:** The proportion of actual late orders successfully identified by the model.
- **F1-score:** A combined measure of precision and recall.

- **ROC-AUC:** A measure of the model's ability to distinguish between the two classes across classification thresholds.

These metrics provide complementary information. In particular, recall is relevant when the objective is to identify as many late orders as possible, while precision indicates how many flagged orders are genuinely late.

6.3.2 Logistic Regression and Random Forest Results

The Logistic Regression model achieved an accuracy of approximately 66.62% and a ROC-AUC of 0.706. Its precision for the late class was approximately 12%, with a recall of 64% and an F1-score of 0.21.

The Random Forest model achieved an accuracy of approximately 78.11% and a ROC-AUC of 0.774. Its late-class precision was approximately 18%, recall was 63%, and F1-score was 0.28.

Metric	Logistic Regression	Random Forest
Accuracy	66.62%	78.11%
ROC-AUC	0.706	0.774
Late-class precision	12%	18%
Late-class recall	64%	63%
Late-class F1-score	0.21	0.28

Table 13. Predictive Model Performance Comparison

The results show differences in the models' classification performance. Random Forest achieved higher accuracy, ROC-AUC, precision, and F1-score in this evaluation, while recall

was similar between the two models. These results describe performance on the selected test set and do not guarantee equivalent performance on future orders.

6.3.3 Interpreting the Results

The relatively low precision for the late class means that many orders flagged as potentially late were not actually late. This is important for practical use: a flagged order should be interpreted as requiring possible review, rather than as a confirmed future delay.

The recall values indicate that the models identified a substantial proportion of late orders in the test set, but neither model detected every late case. This creates a trade-off between identifying more late deliveries and limiting the number of false alarms.

Overall accuracy should be interpreted carefully because the dataset is imbalanced. A model can classify many non-late orders correctly while still performing less effectively on the smaller late-delivery class.

6.3.4 Summary

The evaluation demonstrates that both models captured some predictive signal in the selected order-level features. Random Forest achieved a ROC-AUC of approximately 0.774 and a late-class recall of approximately 63% at the evaluated classification threshold.

The results support further exploration of delivery-risk classification, but the low precision and retrospective nature of the analysis limit immediate operational interpretation. Further validation would be required before using these predictions to guide real delivery interventions.

6.4 Classification Threshold and Feature Importance

The classification threshold determines how the predicted probabilities are converted into late or non-late classifications. Adjusting this threshold changes the balance between identifying late deliveries and generating false alarms. The project therefore examined several thresholds to understand their potential operational implications.

6.4.1 Classification Threshold Analysis

The Random Forest model was evaluated at thresholds ranging from 0.20 to 0.60. At a threshold of 0.50, approximately 23.64% of orders were flagged as potentially late. The model identified approximately 62.9% of actual late deliveries, while precision was approximately 18%.

At a lower threshold of 0.20, the model flagged 97.12% of orders and achieved 99.4% recall, but precision fell to approximately 6.9%. At a threshold of 0.60, only 10.56% of orders were flagged, with precision increasing to approximately 25.6%, while recall decreased to approximately 39.9%.

These results illustrate the trade-off between capturing more late deliveries and limiting the number of orders flagged for review.

Threshold	Share flagged	Precision	Recall
0.20	97.12%	6.9%	99.4%
0.30	78.42%	8.2%	94.9%
0.40	45.88%	11.8%	79.9%
0.50	23.64%	18.0%	62.9%
0.60	10.56%	25.6%	39.9%

Table 14. Classification Threshold Analysis

6.4.2 Selected Threshold

A threshold of 0.50 was used as an illustrative operating point for the project. At this threshold, the model flagged approximately 23.6% of orders while identifying about 62.9% of actual late deliveries.

This threshold is not presented as an objectively optimal business choice. Its suitability would depend on operational capacity, the cost of reviewing flagged orders, and the relative consequences of missed late deliveries versus false alarms.

Given the observed precision, the predictions should be treated as risk indicators for possible review, not as definitive statements that an order will be late.

6.4.3 Feature Importance

The Random Forest feature-importance analysis identified purchase month as the largest contributor to the model's importance scores, at approximately 0.277. Estimated delivery

duration followed at approximately 0.186, while total freight contributed approximately 0.114 and total order value approximately 0.082.

Customer-state indicators also appeared among the more influential features, including São Paulo (SP) at approximately 0.064 and Rio de Janeiro (RJ) at approximately 0.048. The same-state indicator contributed approximately 0.040.

These importance values describe the model's use of features in its fitted predictions. They do not establish causal relationships. For example, the importance assigned to purchase month does not demonstrate that the month itself causes delivery delays; it may reflect patterns associated with time, geography, or other characteristics represented in the dataset.

6.4.4 Interpretation and Limitations

The threshold analysis shows that model performance depends on how predicted probabilities are translated into operational flags. Lower thresholds increase the share of late orders identified but also increase the volume of orders requiring review. Higher thresholds reduce the number of flags but miss a larger proportion of actual late deliveries.

Feature importance provides additional insight into which variables contributed to the fitted Random Forest model. However, importance scores should not be interpreted as causal effects or as evidence that changing a feature would change delivery outcomes.

The results remain subject to the modeling population, feature definitions, and retrospective evaluation design described in the preceding sections. Prospective use would require validation on future orders and confirmation that every input is available at the time a prediction is generated.

6.5 Predictive Modeling Summary and Limitations

The predictive modeling phase explored whether order-level and geographic characteristics could help identify late deliveries. Logistic Regression and Random Forest were evaluated using a stratified 80/20 train-test split, with performance assessed through accuracy, ROC-AUC, precision, recall, and F1-score.

Random Forest achieved an accuracy of approximately 78.11% and a ROC-AUC of 0.774. At the 0.50 classification threshold, it identified approximately 62.9% of late deliveries, while flagging 23.64% of orders. Its late-class precision was approximately 18%, indicating that many flagged orders were not actually late.

The threshold analysis demonstrated that changing the classification threshold alters the volume of flagged orders and the proportion of late deliveries identified. The selected threshold of 0.50 is therefore an illustrative operating point rather than a universally optimal setting.

6.5.1 Main Limitations

Several limitations should be considered when interpreting the modeling results:

- **Class imbalance:** Late deliveries represented approximately 6.77% of the modeling population, making class-specific metrics important.
- **False positives:** The model's relatively low precision means that flagged orders should be treated as candidates for review, not confirmed delays.
- **Retrospective evaluation:** The model was evaluated on historical completed deliveries. Performance on future orders may differ.
- **Feature availability:** Prospective use requires confirming that every feature is available at the time a prediction is generated.
- **Generalizability:** The model's results apply to the analyzed population and should not automatically be extended to orders excluded from the modeling dataset.
- **Interpretability:** Feature-importance scores describe the fitted model's use of variables; they do not demonstrate causal effects.

6.5.2 Overall Contribution

The modeling phase demonstrates how historical e-commerce data can be used to investigate delivery-risk classification and examine the trade-offs involved in operational flagging.

The results provide a starting point for further validation rather than a ready-to-deploy prediction system. Future work should evaluate the model on temporally separated data, confirm feature availability at prediction time, and assess whether its performance is useful under realistic operational constraints.

7. Power BI Dashboard and Business Intelligence

7.1 Data Model and Dashboard Design

Power BI was used to bring together the project's business analysis into an interactive reporting environment. The dashboard organizes key findings across sales, product categories, delivery performance, customers, and sellers, allowing users to explore different aspects of the Olist dataset.

7.1.1 Data Model

The Power BI model includes the core Olist tables: orders, customers, order items, products, sellers, and product-category translations. These tables were connected through their relevant identifiers to support analysis across transactions, customer characteristics, product information, and seller activity.

The principal relationships include:

- Customers connected to orders through `customer_id`.
- Orders connected to order items through `order_id`.
- Products connected to order items through `product_id`.
- Sellers connected to order items through `seller_id`.
- Product-category translations connected to products through the category field.

These relationships allow measures and visualizations to combine information from different parts of the dataset while preserving the underlying transactional structure.

The delivery-risk predictions were imported as a separate table and kept disconnected from the main model. This distinction is important because the predictions require a deliberate matching and validation process before they can be used to filter or enrich the dashboard's existing order-level analysis.

7.1.2 Dashboard Structure

The dashboard is organized into dedicated pages, each addressing a different business-analysis perspective:

1. **Sales Overview** — summarizes overall sales performance and revenue trends.
2. **Product & Category Intelligence** — examines category-level revenue, sales volume, and product performance.
3. **Orders & Delivery** — presents delivery duration, late-delivery rates, and geographic or temporal patterns.
4. **Customer Overview** — explores customer distribution, spending, and RFM segmentation.
5. **Seller Overview** — examines seller activity, revenue, and geographic distribution.
6. **Navigation Page** — provides a starting point for navigating the report.

This structure separates the analysis into focused pages while allowing users to move between related business questions.

7.1.3 Measures and Interactive Analysis

The dashboard uses measures for key indicators, including total orders, delivered orders, completion rate, product revenue, average order value, delivery duration, late-delivery rate, customer metrics, and seller revenue.

Visuals and filters allow users to examine results across dimensions such as time, customer state, product category, and seller state. This supports exploratory analysis by enabling users to focus on particular periods or groups rather than relying only on platform-wide totals.

7.1.4 Design Considerations

The dashboard's page structure is intended to support a progression from overall performance to more detailed analysis. The Sales Overview provides a high-level entry point, while the remaining pages allow users to investigate product, delivery, customer, and seller dimensions.

The dashboard should be interpreted as a business intelligence layer built on the project's prepared data and defined measures. Differences in metric definitions or filter context can affect displayed values, so KPI names, calculation logic, and analysis populations should remain consistent when comparing results across pages.

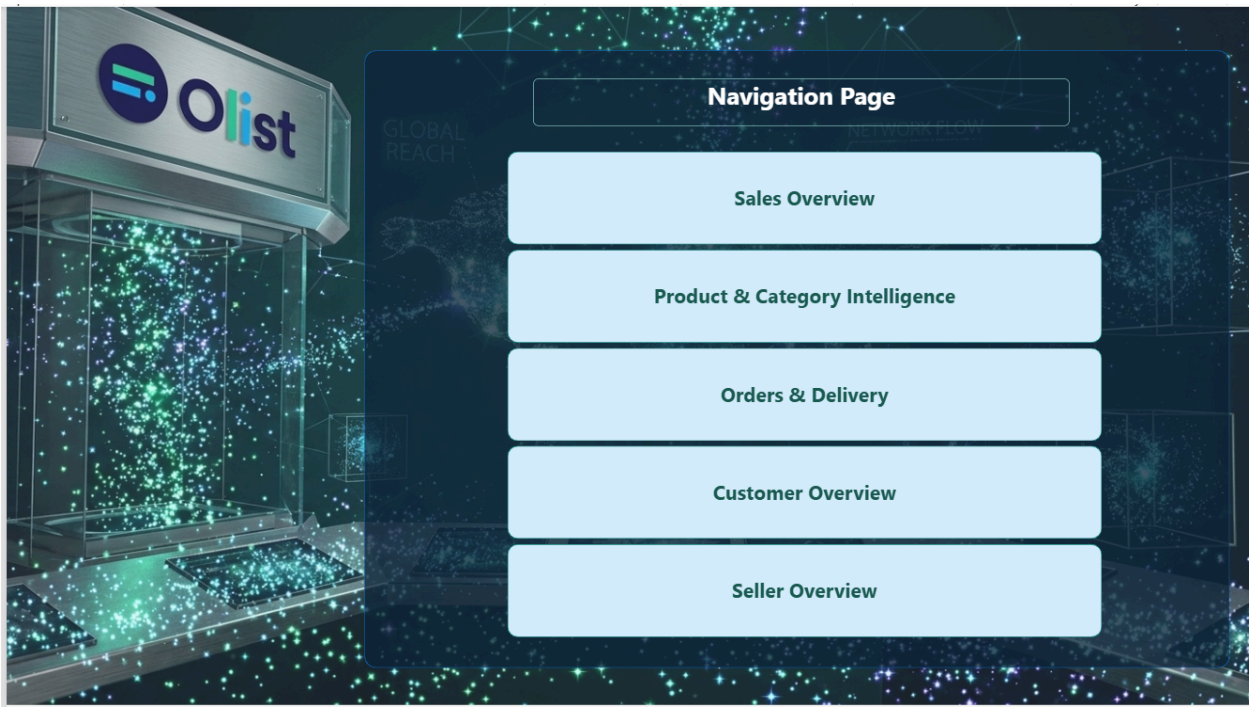


Figure 15. Power BI Dashboards Navigation Pages

7.2 Sales Overview Dashboard

The Sales Overview page provides a high-level view of the platform's commercial performance. It brings together order volume, completed orders, product revenue, average order value, and completion rate to provide a concise overview of the observed e-commerce activity.

7.2.1 Key Performance Indicators

The dashboard reports 99,441 total orders and 96,478 delivered orders, corresponding to a completion rate of 97.02%. Product revenue across the analyzed transactions amounts to approximately €13.59 million.

The dashboard also reports an average order value of approximately €137.04. This measure is based on delivered orders and therefore should be interpreted separately from the transaction-value calculations used in the Python and SQL analyses.

Together, these indicators provide a summary of transaction scale, completed activity, revenue generation, and average order value.

7.2.2 Revenue Evolution

The monthly revenue visualization allows users to examine changes in delivered product revenue over the observed period. The dashboard shows an overall increase in monthly revenue during the dataset period, with the highest observed month occurring in November 2017 at approximately €1.01 million.

The monthly pattern does not indicate a clear, recurring seasonal cycle within the available observation period. Because the dataset covers a limited historical period, this should not be interpreted as evidence that seasonal effects are absent from e-commerce activity more generally.

7.2.3 Geographic Perspective

The Sales Overview page also presents delivered product revenue geographically through a customer-state map. This provides a visual representation of where completed sales were concentrated across Brazil.

The geographic visualization complements the state-level analysis presented earlier by allowing users to explore the distribution of revenue interactively rather than relying only on aggregate totals.

7.2.4 Dashboard Role

The Sales Overview page functions as the main commercial entry point to the dashboard. Its purpose is not to reproduce every analytical result from the project, but to provide a concise starting point from which users can move toward product, delivery, customer, and seller analysis.

The page therefore connects the project’s underlying analytical work with an interactive business intelligence interface, allowing key performance indicators and major sales patterns to be explored within a single view.

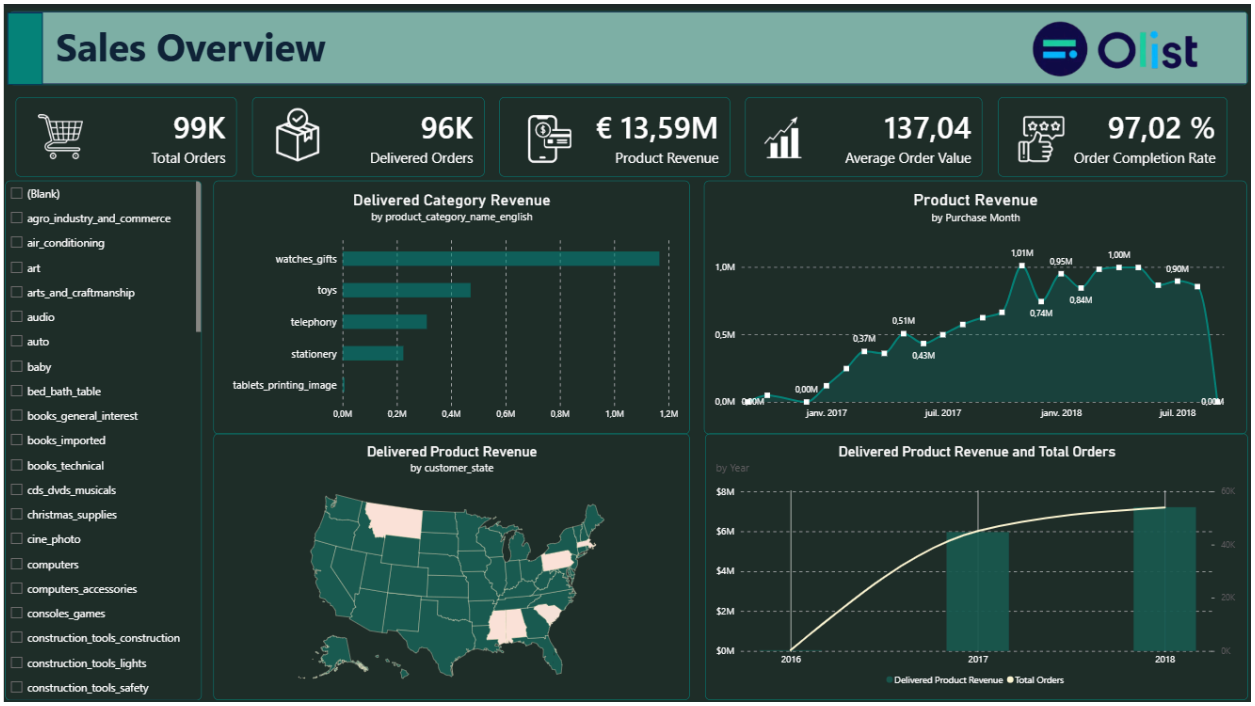


Figure 16. Sales Overview dashboard

7.3 Product & Category Intelligence Dashboard

The Product & Category Intelligence page focuses on the composition of delivered product sales. It combines revenue, units sold, average product price, and category-level comparisons to provide a more detailed view of product performance.

7.3.1 Key Performance Indicators

The page reports approximately €13.22 million in delivered product revenue and approximately 110,197 delivered units sold. The average product price is approximately €120.

These indicators provide three complementary perspectives: the monetary value generated by delivered products, the volume of products sold, and the average price associated with those products.

7.3.2 Category Performance

The dashboard provides comparisons of categories based on delivered product revenue and units sold. The revenue analysis identified health and beauty, watches and gifts, bed, bath and table, sports and leisure, and computers and accessories among the leading revenue-generating categories.

The dashboard also allows categories to be examined according to units sold. This is important because high revenue and high sales volume do not necessarily identify the same categories. A category may generate substantial revenue through a higher average product price, while another may achieve strong revenue primarily through a larger number of units sold.

7.3.3 Product-Level Analysis

The page includes product-level visualizations that complement the category analysis. The top-product comparisons provide a more granular view of which products contribute to revenue and sales volume.

The scatter visualization further allows product performance to be examined through the relationship between product price and sales volume. This provides a way to identify products that combine relatively high prices with substantial sales volume, as well as products with high volume but lower prices.

7.3.4 Dashboard Role

The Product & Category Intelligence page translates the underlying product and category analysis into an interactive business intelligence view. Rather than focusing only on total revenue, it enables users to investigate the composition of that revenue and compare categories and products using multiple performance measures.

This page therefore complements the Sales Overview by moving from the question of **how much was sold** to **what products and categories generated the observed sales**.

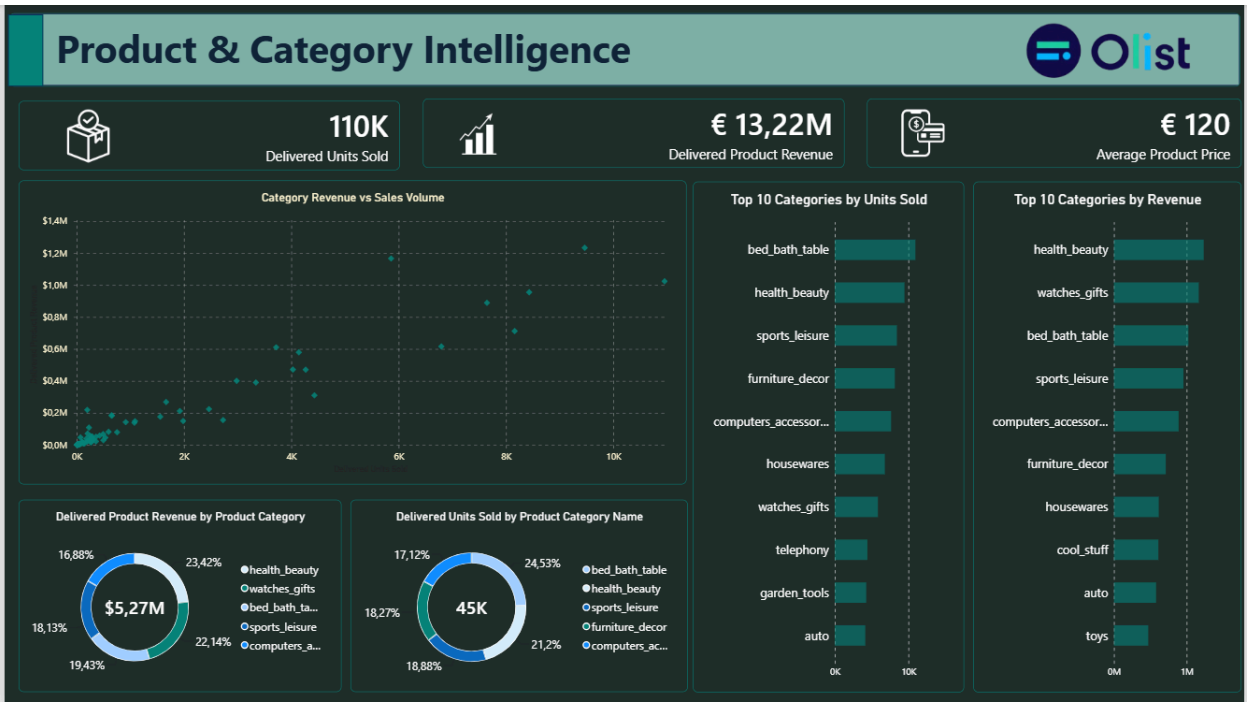


Figure 17. Product & Category Intelligence dashboard

7.4 Orders & Delivery Dashboard

The Orders & Delivery page focuses on operational performance, with particular attention to delivery duration, late deliveries, and geographic differences in delivery outcomes. It translates the delivery analysis developed in Chapter 5 into an interactive monitoring view.

7.4.1 Delivery Performance Indicators

The page reports an average delivery time of approximately 12.5 days and 6,535 late orders based on the dashboard's delivery-status definition. The corresponding late-delivery rate is approximately 6.77%.

These indicators provide a concise view of delivery performance while allowing users to examine the results in greater detail through the page's temporal and geographic visualizations.

7.4.2 Temporal Delivery Patterns

The monthly late-rate visualization allows users to examine how the proportion of late deliveries changes over time. This provides more detail than the overall late-delivery rate by highlighting periods in which the observed rate differs from the overall level.

The dashboard also includes average delivery days by month. This provides a complementary measure because the late-delivery rate and the absolute time required to complete a delivery describe different aspects of delivery performance.

Monthly variations should be interpreted alongside order volume, particularly when a month contains substantially fewer observations than others.

7.4.3 Geographic Delivery Risk

The Delivery Risk by State visualization provides an interactive comparison of delivery outcomes across customer states. This allows users to identify geographic differences in observed delivery risk and to compare these differences with the volume of delivered orders.

As discussed in Chapter 5, state-level late-delivery rates should not be interpreted independently of order volume. Smaller states can display relatively high rates based on fewer observations, while larger states may account for more late orders despite having lower rates.

7.4.4 Dashboard Role

The Orders & Delivery page transforms the delivery analysis into an operational monitoring interface. It brings together overall delivery performance, temporal variation, and geographic differences in a single page.

This page also provides the foundation for the project's subsequent predictive analysis. The descriptive delivery indicators identify the patterns that motivate the question addressed by the machine-learning models: whether historical order characteristics can help identify orders that may be at greater risk of late delivery.

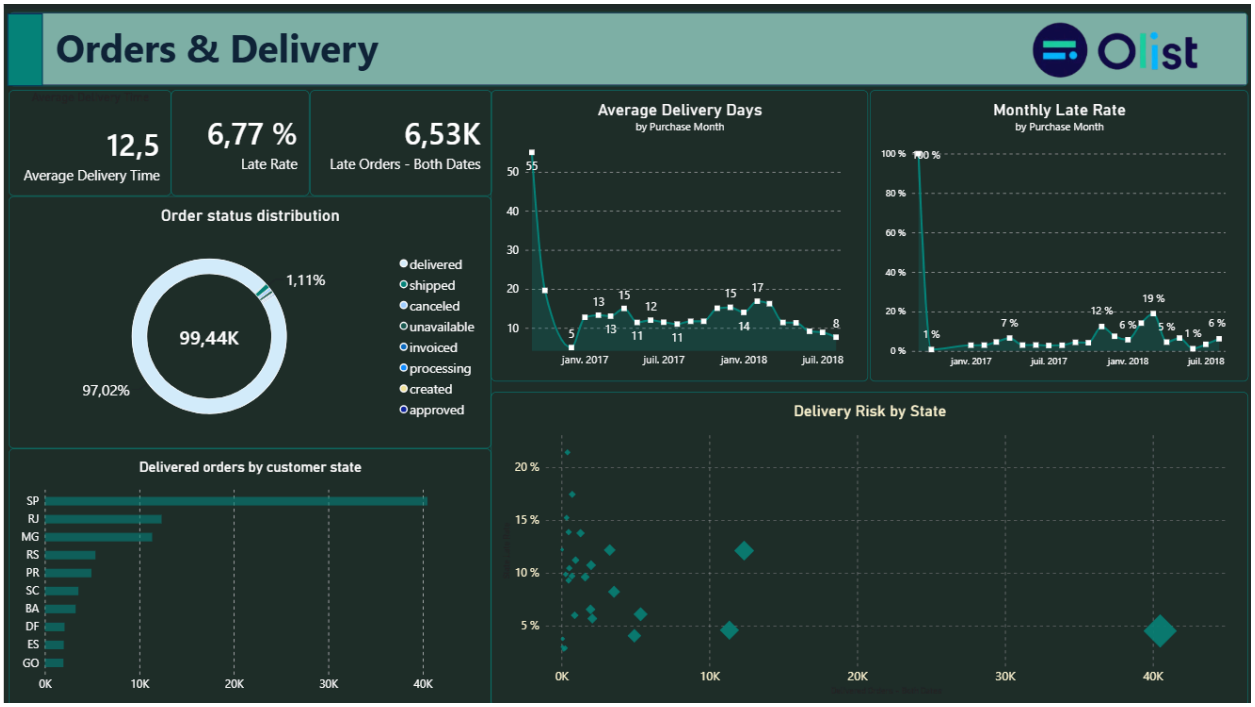


Figure 18. Orders & Delivery dashboard

7.5 Customer Overview Dashboard

The Customer Overview page focuses on customer composition, purchasing behavior, and RFM-based segmentation. It provides an interactive view of the customer-level analysis developed in Chapter 4.

7.5.1 Customer Base and Repeat Purchasing

The dashboard reports approximately 93,000 unique customers within the analytical population and a repeat customer rate of approximately 3.00%.

The low repeat-purchase rate provides important context for interpreting the customer segmentation results. Most observed customers are associated with a limited number of purchases, while a smaller group generates repeated transactions.

The dashboard allows these customer patterns to be examined alongside geographic distribution and customer spending measures.

7.5.2 RFM Customer Segmentation

The Customer Overview page presents the distribution of the RFM segments identified in the analysis. The largest segments are **Other**, representing 43.36% of the RFM customer population, and **Potential Loyalists**, representing 40.06%. **Hibernating / Low Value** customers account for 16.47%, while At Risk, Champions, and Loyal Customers represent very small proportions of the population.

The segment distribution provides an accessible summary of the RFM analysis and allows users to examine the composition of the customer base interactively.

7.5.3 Customer Value

The dashboard also presents average customer spending by segment and revenue contribution by segment. These views complement the customer-count distribution by showing that the relative size of a segment does not necessarily correspond to its contribution to revenue.

For example, the Other segment represents 43.36% of customers but contributes approximately 53.09% of revenue in the RFM analysis. Potential Loyalists represent 40.06% of customers and contribute approximately 40.95% of revenue, while Hibernating / Low Value customers represent 16.47% of customers but contribute approximately 5.56% of revenue.

This comparison demonstrates the value of combining customer counts with monetary measures when interpreting segmentation results.

7.5.4 Geographic Customer Distribution

The customer-state map provides an additional geographic perspective by showing where customers are distributed across Brazil. This allows users to explore the geographic concentration of the customer base and relate customer activity to the geographic patterns observed in the sales and delivery analyses.

7.5.5 Dashboard Role

The Customer Overview page brings together customer scale, repeat purchasing, monetary value, segmentation, and geographic distribution. It therefore provides a customer-centric complement to the transaction and delivery perspectives presented on the preceding dashboard pages.

The page also illustrates how the analytical results developed through Python and SQL can be translated into an interactive interface for exploring customer behavior.

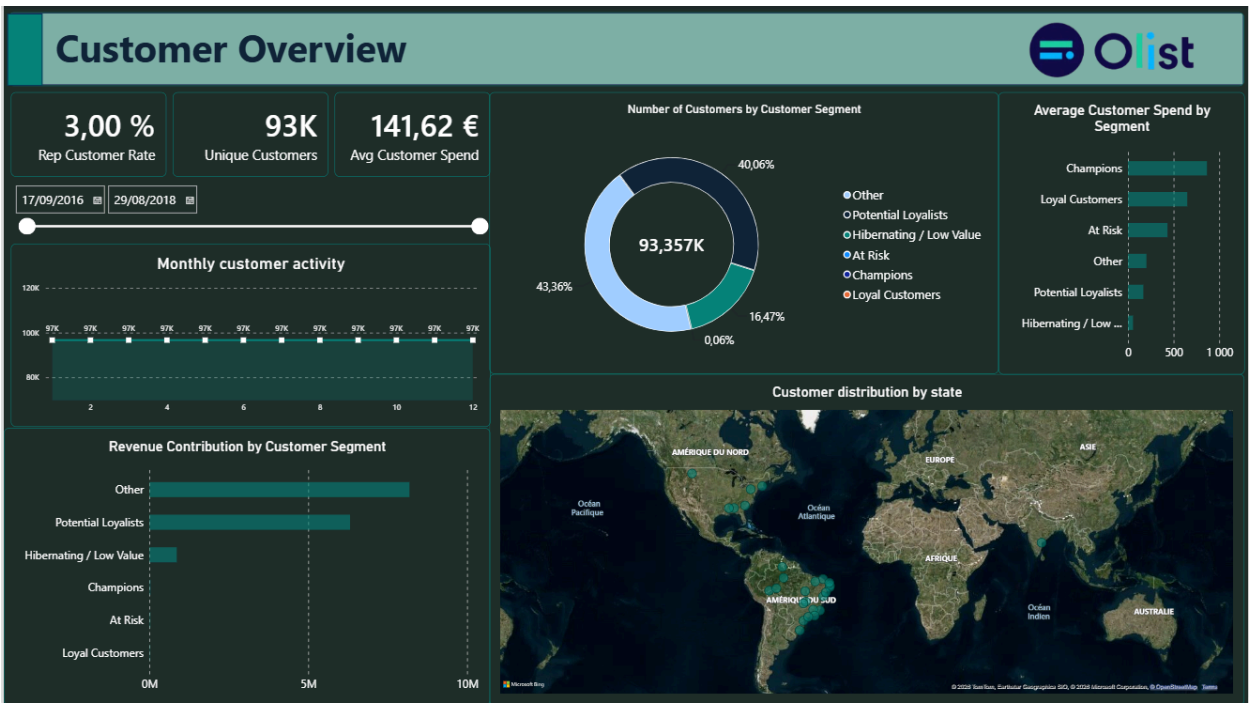


Figure 19. Customer Overview dashboard

7.6 Seller Overview Dashboard

The Seller Overview page provides a seller-centric perspective on the Olist marketplace. It focuses on seller scale, delivered revenue, geographic distribution, and differences in seller activity.

7.6.1 Seller Performance Indicators

The dashboard reports approximately 3,000 sellers and approximately €13.22 million in delivered seller revenue. These indicators provide a high-level view of the size of the seller network and the monetary value associated with completed transactions.

Seller performance is examined at the seller level rather than only through aggregate platform revenue. This allows the analysis to consider differences in seller contribution and geographic distribution.

7.6.2 Geographic Distribution of Sellers

The seller map provides an overview of the geographic distribution of sellers across Brazil. This complements the customer-state analysis by showing the supply-side distribution of the marketplace.

Comparing seller and customer geography can provide context for understanding the spatial structure of the platform, although the dashboard alone does not establish whether geographic proximity directly affects delivery performance.

7.6.3 Seller Revenue Analysis

The dashboard includes a revenue visualization for the leading sellers, allowing users to compare seller contributions to delivered revenue.

This view complements the SQL-based seller concentration analysis presented earlier. The analysis found that the highest-revenue 10% of sellers accounted for 41.10% of revenue, while the highest-revenue 50% accounted for 83.31%.

These figures indicate that seller contributions are unevenly distributed. However, revenue concentration should not automatically be interpreted as operational dependency because the dataset does not measure product substitutability, seller replacement capacity, or the consequences of individual seller disruption.

7.6.4 Dashboard Role

The Seller Overview page provides the supply-side counterpart to the Customer Overview page. While the customer page focuses on demand and purchasing behavior, the seller page examines the marketplace's network of suppliers and their contribution to delivered revenue.

Together, these perspectives provide a broader view of the platform's commercial structure, connecting customer demand with the sellers responsible for fulfilling orders.

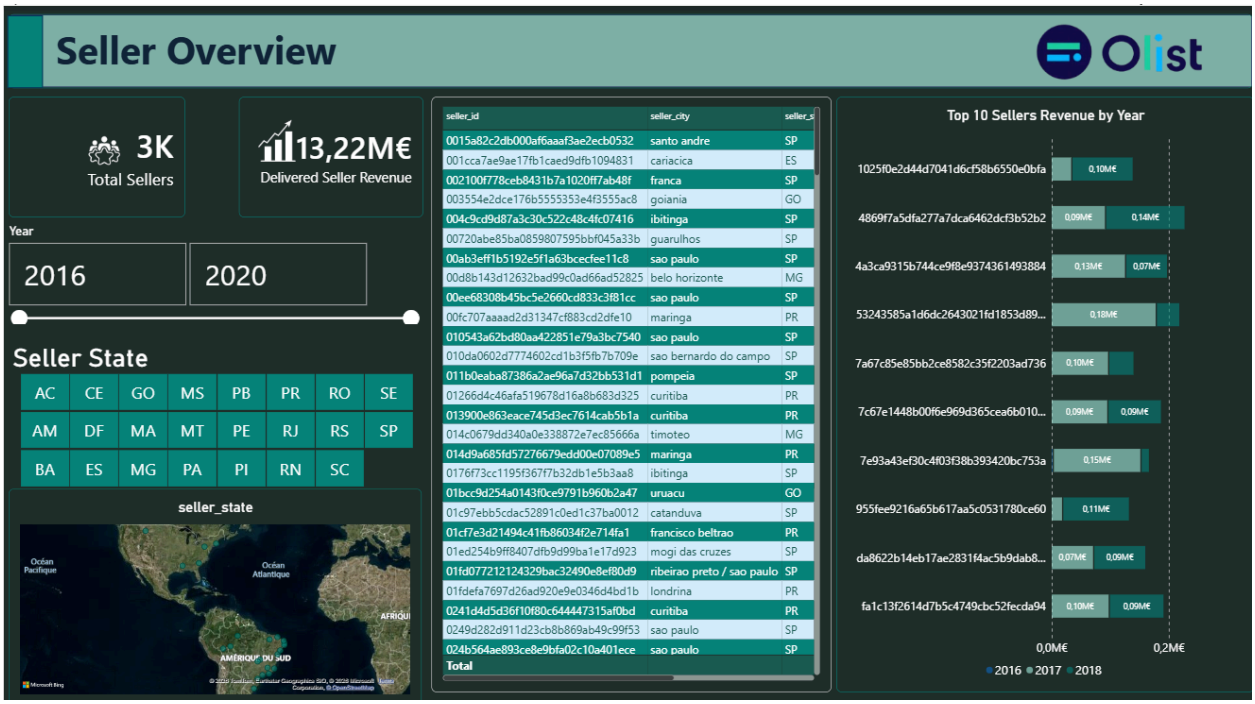


Figure 20. Seller Overview dashboard

7.7 Dashboard Usability and Analytical Limitations

The Power BI dashboard was designed to transform the analytical results obtained through Python and SQL into an interactive business intelligence environment. Across its five analytical pages, users can move from high-level business performance to more specific analyses of products, deliveries, customers, and sellers. The navigation page provides a centralized entry point to these views, while slicers and interactive visual elements allow users to explore the data according to different dimensions such as time, geography, customer segments, and seller characteristics.

A key usability feature of the dashboard is the separation of analytical perspectives. The **Sales Overview** provides a high-level view of orders, revenue, completion, and average order value. The **Product & Category Intelligence** page focuses on product-level and category-level performance. The **Orders & Delivery** page brings together delivery duration, late orders, late rates, and geographic patterns. The **Customer Overview** focuses on customer distribution, repeat purchasing, and RFM segments, while the **Seller Overview** provides visibility into seller activity and revenue concentration. This structure reduces the need to interpret all indicators simultaneously and allows each business question to be investigated within a dedicated analytical context.

The dashboard also improves accessibility of the analytical results by combining KPI cards, trend visualizations, categorical comparisons, geographic representations, and interactive filters. Rather than requiring users to reproduce the Python or SQL analysis, the dashboard presents the main findings in a format suitable for exploratory business analysis and management-level review.

However, several limitations should be considered when interpreting the dashboard. First, some indicators depend on the population and filters used in their calculation. For example, delivery metrics may consider only delivered orders for which the relevant timestamps are available, whereas sales indicators can be calculated over a broader population. Consequently, KPI values should always be interpreted together with their underlying definitions rather than compared without considering their calculation scope.

Second, the dashboard is primarily descriptive and diagnostic. It summarizes historical performance and identifies patterns in sales, customers, products, sellers, and deliveries, but it does not by itself establish causal relationships. This is particularly important for the weather analysis, where differences in delivery outcomes between rainy and non-rainy conditions indicate an association rather than evidence that rainfall directly caused delivery delays.

Third, the predictive delivery-risk dataset is intentionally maintained separately from the main relational model. The model predictions were generated using a retrospective machine-learning workflow and are therefore better interpreted as an analytical demonstration of how delivery-risk scoring could support operational monitoring. The model would require prospective validation on future orders before being used as a production decision-support system.

Finally, the dashboard reflects the characteristics and limitations of the Olist public dataset. The data represents a historical Brazilian e-commerce marketplace and does not necessarily reflect the operational processes, customer behavior, or logistics environment of a specific current company. The dashboard should therefore be viewed as an analytical framework demonstrating how transactional data can be transformed into business insights, rather than as a direct representation of a current company's performance.

Overall, the Power BI implementation provides an interactive layer connecting the project's data preparation, SQL analysis, statistical exploration, external API integration, customer segmentation, and predictive modeling. Its main value lies in bringing these analytical components together into a coherent decision-support interface while maintaining the distinction between descriptive findings, analytical associations, and predictive results.

8. Integrated Insights and Business Recommendations

8.1 Integrated Analytical Findings

The analysis conducted throughout this project reveals several complementary patterns across sales performance, customers, products, sellers, and delivery operations. Rather than interpreting these dimensions independently, their combination provides a broader view of the e-commerce marketplace and highlights areas where operational and commercial actions could be concentrated.

First, the platform demonstrates strong overall order completion, with **96,478 delivered orders out of 99,441 total orders**, corresponding to a completion rate of approximately **97.02%**. Product revenue reached approximately **€13.59 million**, while freight represented an additional **€2.25 million**. Monthly revenue increased substantially during the observed period, reaching its highest level in **November 2017 at approximately €1.01 million**. However, the relatively limited historical period does not provide sufficient evidence to establish a stable recurring seasonal pattern.

Second, the customer analysis highlights a substantial gap between transaction volume and customer retention. Approximately **96,096 unique customers** are represented in the original dataset, while the RFM analysis identified **93,358 customers** with the required purchasing information. Only approximately **3%** of customers were classified as repeat purchasers. At the same time, repeat customers generated a considerably higher average order value than one-time customers, approximately **€260.05 compared with €137.96** in the SQL analysis. This indicates that customer retention represents an important analytical dimension: increasing repeat purchasing could potentially have a meaningful effect on customer value, although the analysis does not estimate the causal impact of specific retention actions.

The RFM analysis further demonstrates that monetary value and customer loyalty should not be treated as equivalent concepts. The largest segment, **Other**, represented approximately **43.36% of customers and 53.09% of revenue**, while **Potential Loyalists** represented approximately **40.06% of customers and 40.95% of revenue**. The small Champions and Loyal Customer groups displayed substantially higher average spending, but represented only a very small proportion of the customer base. This suggests that customer management should differentiate between high-value customers, customers with potential for repeat purchasing, and customers showing limited recent engagement.

Product analysis shows that revenue is distributed across several major categories rather than being generated by a single dominant product group. **Health & Beauty, Watches & Gifts, Bed & Bath Table, Sports & Leisure, and Computers & Accessories** were among the leading categories by delivered product revenue. The Bed & Bath Table category also recorded the

highest unit volume among these leading categories. These results provide a basis for category-level monitoring rather than relying exclusively on overall revenue.

The seller analysis indicates a more concentrated revenue structure. The top **10% of sellers generated approximately 41.10% of revenue**, while the top 20% generated approximately 56.62%. By the top 50%, the cumulative share reached approximately 83.31%. This concentration suggests that seller performance can have a substantial effect on marketplace revenue and that monitoring seller contribution and concentration can complement aggregate sales analysis.

Delivery analysis reveals that approximately **6.77% of delivered orders with both required dates were classified as late**, while most orders were delivered early relative to their estimated dates. The average delivery duration was approximately 12 days in the Python analysis. Late deliveries were associated with substantially longer delivery times, with the SQL analysis showing an average of approximately **33.91 days for late orders compared with 10.82 days for early orders**. The geographic and temporal analyses also indicate that delivery performance varies across locations and periods.

The weather integration provided an additional operational perspective. Among the matched customer orders, the late-delivery rate was approximately **5.89% on days without rain compared with 7.41% on rainy days**, a difference of approximately 1.53 percentage points. Increasing rain intensity was also associated with higher observed late-delivery rates. Nevertheless, these findings should be interpreted as associations rather than causal effects, particularly because the weather variable was based on the actual delivery date and therefore should not be used directly as a predictive feature for prospective delivery-risk modeling.

Finally, the predictive modeling stage demonstrated that historical order characteristics can provide useful signals for identifying potential delivery risk. The Random Forest model achieved an **AUC of approximately 0.774**, compared with approximately 0.706 for Logistic Regression. At the illustrative 0.50 classification threshold, the Random Forest identified approximately **23.6% of orders for review and captured 62.9% of actual late deliveries**. However, its precision for the late class was approximately 18%, meaning that many flagged orders would not ultimately be late. The model should therefore be interpreted as a prioritization mechanism rather than a definitive prediction system.

Taken together, these findings show that the marketplace's analytical priorities extend beyond increasing sales volume. Customer retention, seller concentration, category performance, delivery reliability, and early identification of operational risk represent interconnected dimensions that can be monitored through the analytical framework developed in this project.

8.2 Business Recommendations

The findings provide a basis for several practical recommendations across customer management, delivery operations, product management, and seller performance. These recommendations are derived from the observed patterns in the dataset and should be considered as analytical priorities rather than guaranteed outcomes.

8.2.1 Strengthen Customer Retention

The low repeat-customer rate represents one of the clearest opportunities identified by the analysis. Since repeat customers displayed a substantially higher average order value than one-time customers, the marketplace could place greater emphasis on converting existing customers into repeat purchasers.

A retention strategy could prioritize customers classified as **Potential Loyalists**, since this segment represents a large proportion of the customer base and revenue. Targeted approaches could include personalized product recommendations, post-purchase communication, complementary-product offers, and incentives designed around a second purchase rather than indiscriminate discounts.

High-value segments such as **Champions** and **Loyal Customers** could instead receive differentiated retention treatment focused on maintaining engagement and protecting their existing value. The objective should be to avoid treating all customers identically when their purchasing behavior differs substantially.

8.2.2 Establish Delivery-Risk Monitoring

The delivery analysis and predictive model suggest that delivery reliability should be monitored proactively. A practical implementation would involve creating a delivery-risk monitoring process in which orders are assigned a risk score shortly after purchase and high-risk orders are prioritized for operational review.

The Random Forest model provides a starting point for this approach. At the selected 0.50 threshold, approximately 23.6% of orders were flagged while approximately 62.9% of actual late orders were captured. Because precision was only approximately 18%, the prediction should not be interpreted as a statement that an individual order will necessarily be late. Instead, the score can be used to prioritize operational attention.

Before deployment, the model should be retrained and validated using future orders rather than the historical dataset alone. Its performance should also be monitored over time to determine whether the relationship between order characteristics and delivery outcomes remains stable.

8.2.3 Monitor Delivery Performance by Geography and Time

The observed variation in delivery performance across states and periods indicates that aggregate delivery metrics may conceal localized operational problems. Management could therefore monitor late-delivery rates, average delivery duration, and order volume at geographic and monthly levels.

A monitoring framework could trigger investigation when a region experiences a sustained increase in late-delivery rates or delivery duration. Such monitoring would allow operational teams to distinguish isolated fluctuations from persistent patterns.

Weather information could also be incorporated into operational monitoring as contextual information. The observed association between rainfall and late deliveries suggests that adverse weather conditions may coincide with higher delivery risk. However, weather should be treated as contextual operational information rather than as evidence of a direct causal effect.

8.2.4 Manage Seller Concentration

The strong concentration of revenue among high-performing sellers suggests that seller management should consider both individual performance and marketplace concentration.

The platform could monitor seller-level revenue, order volume, delivery performance, and contribution to total marketplace revenue. High-contributing sellers may require particular attention because disruptions affecting a relatively small group of sellers could potentially affect a substantial share of marketplace activity.

At the same time, concentration metrics should be monitored over time to determine whether dependence on a limited group of sellers is increasing or decreasing.

8.2.5 Use Category-Level Performance for Product Decisions

The category analysis demonstrates that different product categories contribute differently to revenue and sales volume. Product management could therefore combine revenue, unit volume, and average price when evaluating categories rather than relying on a single metric.

High-revenue categories can be monitored for continued performance, while categories with strong unit volume but comparatively lower revenue may require different pricing, assortment, or cross-selling strategies. Category performance should also be evaluated over time to identify sustained changes rather than responding to individual monthly fluctuations.

8.2.6 Establish a KPI Monitoring Framework

The Power BI dashboard provides a foundation for recurring performance monitoring. A practical KPI framework could include:

Business Area	KPI	Purpose
Sales	Total orders	Monitor marketplace activity
Sales	Product revenue	Track commercial performance
Sales	Average order value	Monitor transaction value
Customers	Repeat customer rate	Monitor retention
Customers	Average customer spend	Monitor customer value
Customers	RFM segment distribution	Track customer composition
Delivery	Late-delivery rate	Monitor service reliability
Delivery	Average delivery days	Monitor fulfillment speed
Sellers	Seller revenue contribution	Monitor marketplace concentration
Products	Category revenue	Monitor product performance
Risk	Percentage of orders flagged	Monitor predictive workload

Risk	Recall of late orders	Evaluate risk-screening coverage

Table 15. Business Area and Monitoring KPIs

The KPIs should be reviewed together rather than in isolation. For example, an increase in revenue accompanied by a decline in repeat purchasing or delivery reliability would provide a different managerial signal from revenue growth accompanied by stable customer retention and delivery performance.

Overall, the recommended approach is to combine **customer retention monitoring, delivery-risk management, seller concentration analysis, category performance tracking, and recurring KPI review** into a unified business intelligence process. The Power BI dashboard developed in this project can serve as the reporting layer for this framework, while the Python and SQL workflows provide the analytical foundation behind the indicators.

8.3 Implementation Priorities and Monitoring Framework

The recommendations identified above can be translated into a practical monitoring framework by linking each analytical finding to a measurable indicator and a corresponding management action. This approach helps ensure that the project does not remain limited to descriptive analysis but can support continuous business monitoring.

The first priority should be **customer retention**. The repeat-customer rate should be monitored regularly alongside average order value and the distribution of RFM segments. Changes in the size of the Potential Loyalist, Champions, Loyal, and At Risk groups can provide an early indication of changes in customer engagement. Because repeat customers displayed a higher average order value than one-time customers in the analyzed data, customer retention should remain a central commercial KPI.

The second priority should be **delivery reliability**. The late-delivery rate and average delivery duration should be monitored by month and customer geography. Monitoring these indicators at a more granular level can help identify periods or regions requiring operational investigation. The predictive risk score can complement these indicators by identifying orders that warrant additional attention before the delivery outcome is known.

The third priority should be **seller performance and concentration**. Seller revenue contribution should be tracked alongside order volume and delivery performance. In addition to identifying high-performing sellers, monitoring concentration over time can reveal whether marketplace revenue is becoming increasingly dependent on a smaller group of sellers.

The fourth priority should be **product and category performance**. Revenue, units sold, and average product price should be evaluated together. This allows management to distinguish between categories generating high revenue through large transaction volumes and categories generating high revenue through relatively higher-priced products.

These priorities can be organized into a recurring monitoring cycle:

Priority	Main Indicators	Analytical Action
Customer retention	Repeat rate, RFM segments, average spend	Identify retention opportunities
Delivery reliability	Late rate, delivery days	Detect operational deterioration
Delivery risk	Risk score, flagged orders, recall	Prioritize operational review
Seller performance	Revenue, orders, concentration	Monitor seller dependency
Product performance	Revenue, units, average price	Evaluate category dynamics
Overall performance	Orders, revenue, AOV	Track marketplace evolution

Table 16. Implementation Priorities

The Power BI dashboard can serve as the main monitoring interface, while the underlying SQL queries and Python notebooks can be rerun periodically as new data becomes available. This separation between data processing, analytical modeling, and visualization also makes the workflow easier to update without redesigning the entire dashboard.

For the predictive component specifically, implementation should follow a validation cycle before operational adoption. New orders should be scored using a model trained exclusively on historical information available before the prediction point. Actual delivery outcomes can then be compared with the predictions to calculate precision, recall, and other performance indicators. This process would determine whether the model remains sufficiently reliable when applied to future data.

Overall, the proposed framework establishes a continuous link between **data collection, analytical processing, visualization, business interpretation, and operational monitoring**. It provides a practical extension of the analytical workflow developed in this project and creates a structure that could be adapted to a larger or continuously updated e-commerce environment.

9. Conclusion and Future Work

9.1 Conclusion

This project developed an end-to-end business intelligence and analytics framework using the Olist Brazilian E-Commerce Public Dataset. By combining Python-based data preparation and exploratory analysis, SQL-based querying, external weather data integration, customer segmentation, predictive modeling, and Power BI visualization, the project transformed raw transactional data into a structured analytical environment.

The analysis provided insights into several dimensions of marketplace performance. Sales analysis highlighted revenue evolution and category-level contributions, while customer analytics revealed a low repeat-purchase rate and substantial differences in customer spending behavior. Seller analysis identified a concentrated revenue structure, and delivery analysis highlighted variations in delivery duration and late-delivery rates across orders and locations.

The integration of historical weather data added an operational perspective by identifying an association between rainfall and late deliveries. The predictive modeling stage further demonstrated how order characteristics could be used to identify potential delivery risks, although the model's limited precision and retrospective evaluation mean that it should not be considered ready for direct operational deployment.

The Power BI dashboard brought these analytical components together into an interactive reporting environment, enabling users to explore sales, products, customers, sellers, and delivery performance through dedicated analytical views.

Overall, the project demonstrates how a structured analytics workflow can connect data preparation, statistical exploration, customer analysis, predictive techniques, and business intelligence visualization. Its main contribution is the development of a coherent framework that supports evidence-based interpretation of e-commerce performance while acknowledging the limitations of historical data and predictive uncertainty.

9.2 Limitations and Future Work

Despite its analytical scope, the project presents several limitations.

First, the dataset represents a historical period and a specific Brazilian e-commerce marketplace. The findings cannot automatically be generalized to other markets, platforms, or current operating conditions.

Second, the analysis depends on the completeness and consistency of the available transactional data. Missing timestamps, differences in analytical populations, and the use of different calculation methods can affect KPI comparability.

Third, the weather analysis identifies associations rather than causal relationships. A more rigorous investigation would require additional contextual variables and a design capable of separating weather effects from other operational factors.

Fourth, the predictive model was evaluated retrospectively using a train-test split. Future work should include temporal validation, calibration, additional performance evaluation, and testing on genuinely future orders before considering operational deployment.

Several extensions could strengthen the project:

- **Customer analytics:** Develop more detailed retention and customer-lifetime-value analyses, with a focus on understanding repeat purchasing over time.
- **Delivery analytics:** Investigate additional factors associated with delays, including logistics characteristics and geographic distance where reliable data is available.
- **Predictive modeling:** Compare additional models, evaluate temporal generalization, and refine the classification threshold according to operational costs and available review capacity.
- **Weather analysis:** Incorporate weather forecasts available before delivery and assess whether they provide useful predictive information without introducing information leakage.
- **Business intelligence:** Automate data refreshes and establish a recurring KPI monitoring process to support analysis as new data becomes available.

These extensions would move the project from a retrospective analytical study toward a more dynamic decision-support framework.

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Appendix A — Dataset and Data Dictionary

A.1 Dataset Overview

The project uses the Brazilian E-Commerce Public Dataset by Olist, which contains transactional information about orders, customers, products, sellers, payments, and reviews. The dataset was organized into relational tables and processed using Python and MySQL before being integrated into Power BI.

The principal tables used in the project are summarized below.

Table	Description	Main Analytical Use
`orders`	Order records and order-related timestamps and statuses	Sales and delivery analysis
`order_items`	Products purchased within each order, including prices and freight values	Revenue and product analysis
`customers`	Customer records and geographic information	Customer and geographic analysis
`products`	Product attributes and category information	Product and category analysis
`sellers`	Seller records and geographic information	Seller performance analysis
`category_translation`	Translation of product categories into English	Category reporting

customer_rfm	Derived customer-level recency, frequency, and monetary metrics	Customer segmentation
delivery_risk_predictions	Model-generated delivery-risk scores and predictions	Predictive risk analysis

Table 17. Dataset tables Definition

A.2 Main Fields and Definitions

Orders

Field	Description
order_id	Unique order identifier
customer_id	Customer identifier associated with the order
order_status	Recorded status of the order
order_purchase_timestamp	Date and time when the order was placed
order_approved_at	Timestamp of payment approval
order_delivered_carrier_date	Date the order was handed to the carrier

`order_delivered_customer_date`	Date the order was delivered to the customer
`order_estimated_delivery_date`	Estimated delivery date

Table 18. Orders Table columns definitions

Order Items

Field	Description
`order_id`	Identifier of the associated order
`order_item_id`	Item sequence within the order
`product_id`	Identifier of the purchased product
`seller_id`	Identifier of the seller
`price`	Product price
`freight_value`	Freight charge associated with the item

Table 19. Order_items table columns definitions

Customers

Field	Description
`customer_id`	Customer record identifier
`customer_unique_id`	Identifier used to distinguish unique customers
`customer_city`	Customer city
`customer_state`	Customer state

Table 20. Customer Table columns definitions

Products

Field	Description
`product_id`	Unique product identifier
`product_category_name`	Product category in the original dataset
Product attribute fields	Additional product characteristics, including dimensions and weight where available

Table 21. Products Table columns definitions

Sellers

Field	Description
`seller_id`	Unique seller identifier
`seller_city`	Seller city
`seller_state`	Seller state

Table 22. Sellers Table columns definitions

A.3 Derived Analytical Tables

The project also includes derived tables created during the analytical workflow.

Table	Main Contents
`customer_rfm`	Customer recency, purchase frequency, spending metrics, RFM scores, and segment assignments
`delivery_risk_predictions`	Predicted delivery-risk probabilities, classifications, and associated order-level information

Table 23. Derived Analytical Tables

These derived tables support the customer segmentation and predictive modeling stages and extend the original transactional dataset with analytical outputs.

Note: This appendix summarizes the principal fields used in the project. It is not an exhaustive schema of every original dataset column or every intermediate Python and SQL table.

Appendix B — SQL Queries and KPI Definitions

B.1 SQL Analysis Overview

SQL was used to query the relational dataset stored in the MySQL database `olist_analytics`. The analysis focused on order performance, revenue, customer purchasing behavior, delivery outcomes, and seller contribution.

The queries relied primarily on the `orders`, `order_items`, `customers`, `products`, and `sellers` tables. Aggregations and joins were used to calculate business indicators and prepare analytical results for comparison with the Python analysis and Power BI dashboard.

B.2 Main KPI Definitions

KPI	Definition	Analytical Purpose
Total Orders	Count of distinct order identifiers	Measures order volume
Delivered Orders	Count of distinct orders with delivered status	Measures completed deliveries
Completion Rate	Delivered orders divided by total orders	Measures the share of orders delivered
Product Revenue	Sum of product prices across order items	Measures product sales value

Freight Revenue	Sum of freight charges across order items	Measures shipping charges
Total Transaction Value	Product revenue plus freight revenue	Measures combined product and freight value
Average Order Value	Delivered product revenue divided by delivered orders	Measures average product value per delivered order
Repeat Customer Rate	Customers with more than one purchase divided by the analyzed customer population	Measures repeat purchasing
Late-Delivery Rate	Delivered orders classified as late divided by delivered orders with both required dates	Measures delivery punctuality
Average Delivery Duration	Average number of days between purchase and delivery	Measures fulfillment duration
Seller Revenue Contribution	Seller revenue divided by total revenue	Measures seller-level contribution

Table 24. Main KPI definitions

B.3 Revenue and Order Performance

The SQL analysis calculated order volume, delivered orders, product revenue, freight revenue, total transaction value, and average order value.

The main revenue indicators were:

Indicator	Result
Product Revenue	€13,591,643.70
Freight Revenue	€2,251,909.54
Total Transaction Value	€15,843,553.24
Average Order Value	€137.75

Table 25. Revenue & order Performance

The total transaction value represents the combined value of product prices and freight charges. It should not be interpreted as net profit because the dataset does not provide the complete cost information required for profitability analysis.

B.4 Customer Purchasing Behavior

Customer purchasing behavior was analyzed by grouping transactions according to the unique customer identifier and distinguishing one-time from repeat purchasers.

The SQL results showed the following:

Customer Type	Customers	Share of Customers	Revenue Share	Average Order Value
One-time	90,557	97.00%	94.49%	€137.96
Repeat	2,801	3.00%	5.51%	€260.05

Table 26. Customer Purchasing Behavior

These results highlight the difference in average order value between one-time and repeat customers.

B.5 Delivery Performance

Delivery performance was evaluated using the purchase date, actual delivery date, and estimated delivery date.

The delivery classification distinguished between:

- **Early:** Actual delivery occurred before the estimated delivery date.
- **On Time:** Actual delivery occurred on the estimated delivery date.
- **Late:** Actual delivery occurred after the estimated delivery date.

The late-delivery rate was calculated using only delivered orders for which both the actual and estimated delivery dates were available. This ensures that the denominator corresponds to the population used for the classification.

B.6 Seller Revenue Concentration

Seller revenue concentration was analyzed by calculating seller-level revenue, ranking sellers by revenue, and examining cumulative revenue shares across seller groups.

The analysis showed that the top 10% of sellers generated approximately 41.10% of revenue, while the top 20% generated approximately 56.62%. The top 50% accounted for approximately 83.31%.

These indicators were used to assess the distribution of marketplace revenue across sellers.

Note: This appendix documents the principal SQL indicators and their calculation logic. The exact executable SQL statements should be included separately from the original project notebooks or query files to preserve their actual syntax and filtering conditions.

Appendix C — Python Workflow and Data Preparation

C.1 Overview

Python was used as the primary environment for data preparation, exploratory data analysis, customer segmentation, weather data integration, and predictive modeling. The workflow was

organized into sequential stages, allowing the original transactional data to be transformed into analytical datasets for SQL analysis and Power BI reporting.

The main libraries used included pandas for data manipulation and scikit-learn for machine-learning tasks. Python notebooks were used to document the analytical process and generate intermediate datasets.

C.2 Data Preparation

The data preparation stage involved loading the original Olist datasets, examining their structure, and preparing them for analysis.

The main operations included:

- Inspecting dataset dimensions, data types, and missing values.
- Converting relevant timestamp fields into appropriate date formats.
- Checking identifiers and relationships between tables.
- Merging order, customer, product, and seller information where required.
- Preparing derived datasets for revenue, delivery, and customer-level analysis.
- Exporting processed datasets for subsequent SQL and Power BI use.

Data preparation was performed with attention to preserving the distinction between order-level records and item-level records to avoid inappropriate aggregation.

C.3 Exploratory Data Analysis

Exploratory analysis was used to examine the main characteristics of the marketplace, including:

- Order volume and completion.
- Revenue evolution over time.
- Product and category performance.
- Geographic distribution of customers and sellers.
- Delivery duration and delivery delays.
- Customer purchasing frequency and spending behavior.

The results of this stage informed the KPI definitions, customer segmentation, and subsequent analytical questions.

C.4 Customer Segmentation

Customer segmentation was conducted using the Recency, Frequency, and Monetary (RFM) framework.

The analysis derived customer-level indicators including:

Metric	Definition
Recency	Number of days since the customer's most recent purchase
Frequency	Number of purchases recorded for the customer
Monetary Value	Customer spending based on the selected transaction-value definition

Table 27. Customer Segmentation

Customers were assigned RFM scores and grouped into behavioral segments. The resulting customer-level dataset was exported for further analysis and integration into Power BI.

C.5 Weather Data Integration

Weather data was retrieved using the Open-Meteo API and combined with delivery information.

The workflow involved geocoding customer cities, retrieving historical weather observations, and matching weather records to delivery locations and dates. The resulting dataset was used to compare late-delivery rates across weather conditions.

The analysis was interpreted as an investigation of associations between weather conditions and delivery outcomes, not as a causal assessment.

C.6 Predictive Modeling

Python was also used to develop and evaluate delivery-risk classification models.

The modeling workflow included:

1. Defining the late-delivery target.
2. Selecting order-level predictive features.
3. Splitting the dataset into training and testing subsets using stratification.
4. Training Logistic Regression and Random Forest classifiers.

5. Evaluating model performance using classification metrics and ROC-AUC.
6. Examining classification thresholds and feature importance.
7. Exporting predictions for potential integration into Power BI.

The resulting prediction dataset was used to explore how risk scores could support delivery monitoring. The model remains a retrospective analytical prototype requiring further validation before operational use.

C.7 Outputs

The Python workflow produced processed datasets and analytical outputs supporting the project's SQL queries, customer segmentation, weather analysis, predictive modeling, and Power BI dashboards.

The notebooks serve as the technical record of the transformations, calculations, and modeling procedures used throughout the project.

Note: This appendix summarizes the Python workflow. The complete executable code, exact notebook filenames, and detailed outputs should be included or referenced from the project's source files.

Appendix D — Predictive Modeling Details and Evaluation

D.1 Modeling Objective

The predictive modeling component aimed to identify orders at risk of late delivery using information available at the order level. The objective was to explore whether machine-learning models could support earlier identification of potentially delayed orders and help prioritize operational monitoring.

The target variable was defined using the actual and estimated delivery dates for delivered orders with both dates available.

D.2 Dataset and Target Definition

The modeling dataset contained **96,476 delivered orders** with the required delivery information.

The target variable distinguished between:

- **Late:** Orders delivered after the estimated delivery date.
- **Non-late:** Orders delivered on or before the estimated delivery date.

The target distribution was:

Target Class	Number of Orders	Percentage
Non-late	89,941	93.23%
Late	6,535	6.77%
Total	**96,476**	**100%**

Table 28. Dataset & Target Definition

The target distribution indicates a substantial class imbalance, with late deliveries representing a relatively small proportion of the observations.

D.3 Predictive Features

The models used order-level information selected to represent transaction characteristics, geographic relationships, and estimated delivery timing.

Feature	Description
`total_order_value`	Total monetary value associated with the order
`total_freight`	Total freight charges associated with the order
`number_of_items`	Number of items in the order

`number_of_sellers`	Number of distinct sellers involved in the order
`same_state`	Indicator of whether customer and seller are in the same state
`purchase_month`	Month in which the order was placed
`estimated_delivery_days`	Estimated delivery duration
`customer_state`	Customer's state
`seller_state`	Seller's state

Table 29. Predictive Features Description

These variables were used to investigate whether transaction size, order composition, geography, and estimated delivery timing contained useful signals for classifying late deliveries.

D.4 Training and Evaluation Approach

The dataset was divided into training and testing subsets using an **80/20 stratified split**. Stratification was used to preserve the approximate distribution of late and non-late orders across the two subsets.

Two classification algorithms were evaluated:

1. **Logistic Regression:** A linear classification model used as a baseline.
2. **Random Forest:** An ensemble model combining multiple decision trees to capture potentially nonlinear relationships.

The models were evaluated using accuracy, ROC-AUC, precision, recall, and F1-score. Particular attention was given to the performance of the late-delivery class because of the imbalance between the two target classes.

D.5 Model Performance

The test-set results were:

Metric	Logistic Regression	Random Forest
Accuracy	0.6662	0.7811
ROC-AUC	0.7063	0.7736
Late-Class Precision	0.12	0.18
Late-Class Recall	0.64	0.63
Late-Class F1-Score	0.21	0.28

Table 30. Model Performance

The Random Forest achieved higher accuracy, ROC-AUC, precision, and F1-score than Logistic Regression. Recall for the late-delivery class was similar between the two models.

The results indicate that the Random Forest provided stronger overall classification performance in this evaluation. However, the relatively low precision of both models means that many orders classified as late were not actually late.

D.6 Classification Threshold Analysis

The Random Forest classification threshold was examined to understand the trade-off between the number of orders flagged for review and the proportion of late deliveries identified.

Threshold	Orders Flagged	Precision	Recall
0.20	97.12%	0.069	0.994
0.30	78.42%	0.082	0.949
0.40	45.88%	0.118	0.799
0.50	23.64%	0.180	0.629
0.60	10.56%	0.256	0.399

Table 31. Classification Threshold Analysis

A threshold of **0.50** was selected as an illustrative operating point for the dashboard integration. At this threshold, approximately 23.64% of orders were flagged, while approximately 62.9% of actual late deliveries were identified.

This threshold represents a compromise between the number of orders requiring review and the coverage of late deliveries. It should not be interpreted as an optimized operational threshold because the appropriate balance depends on review capacity and the relative costs of missed delays and unnecessary alerts.

D.7 Feature Importance

The Random Forest feature-importance analysis identified the following variables among the most influential:

Feature	Importance
Purchase Month	0.277
Estimated Delivery Days	0.186
Total Freight	0.114
Total Order Value	0.082
Customer State: SP	0.064
Customer State: RJ	0.048
Same State	0.040

Table 32. Feature Importance

Purchase month and estimated delivery duration had the highest reported importance values. These results indicate that temporal, logistical, and geographic characteristics contributed to the model's classification process.

Feature importance describes how the model uses variables for prediction. It does not establish that these variables directly cause delivery delays.

D.8 Limitations

The modeling results should be interpreted in light of several limitations.

First, the evaluation was retrospective and based on a stratified train-test split rather than a dedicated future-period validation. Second, the target distribution was imbalanced, making accuracy alone insufficient for assessing performance. Third, the low precision of the late-delivery class means that a substantial proportion of flagged orders would not ultimately be late.

The model should therefore be considered an analytical prototype for delivery-risk prioritization. Further temporal validation, calibration, and operational testing would be required before deployment.

Appendix E — Power BI Dashboard Documentation

E.1 Dashboard Overview

The Power BI dashboard was developed to present the main analytical findings of the project through an interactive business intelligence interface. It consists of five analytical pages and a navigation page, each dedicated to a specific dimension of marketplace performance.

The dashboard integrates sales, product, customer, seller, and delivery indicators into a unified reporting environment. Its purpose is to facilitate the exploration of historical performance and make the analytical results accessible through interactive visualizations and filters.

E.2 Dashboard Pages

E.2.1 Sales Overview

The Sales Overview page provides a high-level summary of marketplace performance.

It includes KPI cards for total orders, delivered orders, product revenue, average order value, and completion rate. Additional visualizations examine monthly delivered revenue, category-level revenue, geographic revenue distribution, and annual performance.

This page serves as the main entry point for understanding overall commercial activity.

E.2.2 Product & Category Intelligence

The Product & Category Intelligence page focuses on product-level and category-level performance.

It presents delivered product revenue, units sold, and average product price. Visualizations compare leading categories by revenue and unit volume and explore the relationship between product price and sales performance.

This page supports the identification of categories contributing substantially to marketplace revenue and sales volume.

E.2.3 Orders & Delivery

The Orders & Delivery page focuses on order fulfillment and delivery performance.

Its indicators include average delivery duration, late-order count, and late-delivery rate. Visualizations examine late-delivery trends over time, geographic variation, delivery duration by month, order status distribution, and delivered orders by customer state.

This page provides an operational perspective on delivery reliability and fulfillment performance.

E.2.4 Customer Overview

The Customer Overview page presents customer purchasing behavior and segmentation results.

It includes customer segment distribution, revenue contribution by segment, average spending by segment, repeat-customer rate, and unique-customer count. Geographic and temporal visualizations provide additional context for customer activity.

The page supports the interpretation of customer value and purchasing patterns, particularly through the RFM segmentation framework.

E.2.5 Seller Overview

The Seller Overview page examines seller activity and revenue contribution.

It includes the total number of sellers, delivered seller revenue, geographic distribution, and comparisons of seller revenue across the available analytical dimensions. Filters allow users to explore seller performance by year and seller state.

This page supports the analysis of seller contribution and revenue concentration within the marketplace.

E.2.6 Navigation Page

The Navigation page provides a centralized interface for moving between the dashboard's analytical pages.

Its purpose is to improve accessibility and help users navigate the different perspectives of marketplace performance without having to search through individual report tabs.

E.3 Data Model and Relationships

The Power BI model integrates the principal transactional tables used in the analysis.

The main relationships include:

Primary Table	Related Table	Relationship
Customers	Orders	One-to-many
Orders	Order Items	One-to-many
Products	Order Items	One-to-many
Category Translation	Products	One-to-many
Sellers	Order Items	One-to-many

Table 33. Data Model & Relationships

These relationships allow users to analyze order and item-level information through customer, product, category, and seller dimensions.

The delivery-risk prediction dataset is maintained separately from the main relational model. This reflects its role as a model-generated analytical output rather than an original transactional dimension.

E.4 Dashboard Interactivity

The dashboard uses KPI cards, charts, geographic visualizations, and interactive filters to support exploratory analysis.

Users can examine performance across different analytical dimensions, including time, geography, product categories, customer segments, and sellers, depending on the page.

The combination of dedicated analytical pages and interactive controls allows the dashboard to serve as a reporting interface for the project's main findings.

E.5 Dashboard Scope and Limitations

The dashboard presents historical data and should be interpreted within the scope of the Olist dataset.

Its indicators depend on the definitions and populations used in their calculations. In particular, delivered-order metrics, customer-level indicators, and seller-level revenue measures may use different analytical populations.

The dashboard is primarily descriptive and diagnostic. It does not independently establish causal relationships, and the delivery-risk predictions should be treated as outputs of a retrospective modeling exercise rather than validated operational forecasts.

Overall, the Power BI dashboard consolidates the project's analytical outputs into a structured and interactive reporting environment, connecting the technical workflow to the business questions investigated throughout the report.